

RESEARCH ARTICLE

Modelling and Forecasting of Complex Multiple Seasonalities in Categorical Time Series using TSOLR and ISOLR Models

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ABSTRACT

Multiple seasonality have been widely studied in continuous time series using models such as TBATS (Trigonometric, Box-Cox transform, ARMA errors, Trend and Seasonal components) and MSTL (Multiple Seasonal-Trend decomposition using LOESS). However, their treatment in categorical time series, such as air quality index (AQI) data, remains limited. Categorical AQI often exhibits distinct seasonal patterns at multiple frequencies, which are not captured by standard models. In this paper, we propose a framework that models multiple seasonality using Fourier series and indicator functions. The approach accommodates the ordinal nature of AQI categories while explicitly capturing weekly, monthly and yearly seasonal cycles. Simulation studies demonstrate the empirical consistency of parameter estimates and the forecasting performance of the proposed models under different data generating processes. We further illustrate its applicability using real categorical AQI data from Kolkata and Mumbai and compare forecasting performance with different methods.

1 | Introduction

A categorical time series is a sequence of observations recorded over time where each observation belongs to a finite set of categories rather than numerical values. For example, daily Air Quality Index (AQI) is recorded as Good, Satisfactory, Moderate, Poor, Very Poor and Severe. Hourly traffic conditions can be recorded as Low, Medium or High based on traffic volumes. Analyzing a categorical time series is challenging because elementary arithmetic operations like addition, multiplication or division between two or multiple categories are not defined resulting in statistical summaries like mean, variance and autocovariances are not applicable. One needs to employ models that can bypass these numerical codings. Overall, the prime objective of studying categorical time series data is to provide suitable statistical models

to capture temporal patterns such as trend, seasonality, structural break, extreme event cluster formation or forecasting future cases while retaining the discrete nature of the data. In this paper, we propose to model multiple seasonality for categorical time series with application to daily AQI data.

Seasonality in a time series refers to repeated patterns observed over a period of time where duration of a seasonal cycle is less than a year. For example, air quality is poorer during winter than during summer, indicating yearly seasonality. Similarly, traffic congestion may show high traffic during office hours each day and relatively low traffic during other times, indicating daily seasonality. In some cases, there could be presence of several seasonalities in a single time series. For example, daily AQI data can have weekday versus weekend seasonality, summer ver-

sus winter seasonality and festive versus non-festive seasonality. Modeling such multiple seasonalities for a categorical time series is challenging.

In real-valued time series, where observations are numerical, several methods exist to model multiple seasonality. For example, in case of hourly electricity demand data, multiple seasonality can be modeled through MSTL, TBATS and their extended versions. However, these methods cannot be used for categorical time series, as they involve arithmetic operations, which are not applicable to categorical observations. These limitations, however, can be bypassed by using probability of the categories and then defining features like seasonality on these probabilities.

In mathematical language, a time series process $\{Y_t\}$ is said to have seasonality with seasonal frequency d if $Y_t = Y_{t-d}$ for all t . This implies that the value of Y_t remains the same at every d -time gap. This definition can be extended to categorical time series by using marginal probabilities. A categorical time series Y_t is said to have seasonality of period d if

$$P(Y_t = C_i) = P(Y_{t-d} = C_i)$$

for all t and i . Here, we propose an extension of the TBATS framework tailored for categorical time series using these probabilities. Specifically, we employ a proportional odds ordinal logistic regression model (Agresti 2002) to account for the ordered nature of AQI categories while incorporating Fourier series terms as covariates to flexibly represent multiple seasonal patterns at daily, weekly and annual frequencies. This design preserves the interpretability of seasonal effects, as the Fourier coefficients directly reflect the contribution of each cycle while ensuring consistency with the ordinal data structure.

As an alternative to Fourier-based representations, multiple seasonalities can also be modeled using indicator functions that directly encode recurring time segments. For example, separate dummy variables may be introduced for each hour of the day, day of the week or month of the year. Unlike Fourier series, which approximate seasonal cycles smoothly using sinusoidal functions, indicator functions allow for sharp changes and irregular seasonal shapes that are often observed in practice, such as sudden spikes in AQI levels during specific festival days or winter months. This approach provides more interpretability in representing heterogeneous seasonal effects across categories. Hence, indicator-based modeling offers a complementary perspective to Fourier expansions and can be especially useful when the seasonal structure is highly irregular or difficult to approximate with smooth harmonic terms.

The remainder of this article is structured as follows. Section 2 provides a comprehensive review of existing research on AQI modeling and forecasting techniques. Section 3 describes the dataset and includes an exploratory analysis that motivates the modeling approach. The proposed ordinal logistic regression models are introduced in Section 4. Section 5 details the estimation procedure for the proposed models. Some existing methods for modeling and forecasting categorical time series data are summarized in Section 6. Section 7 discusses the procedure for selecting the order of the model based on various auto-association measures. Simulation experiments assessing the consistency and

forecasting performance of the proposed models are detailed in Section 8. Section 9 presents the AQI data analysis results. Finally, Section 10 concludes the article and suggests directions for future research.

2 | Literature Review

Numerous studies have focused on modeling air quality using time series techniques, with most treating AQI as a continuous variable and applying models such as ARIMA, VAR and state-space approaches (Sethi and Mittal 2020; Liu et al. 2021, 2024; Gupta, Mohta, et al. 2023). These methods are effective for continuous data. But AQI is commonly reported in ordinal categories for public health communication. So these continuous-value models cannot be applied directly to the categorical AQI data. Research on time series models for categorical AQI data remains relatively limited (Liu et al. 2022; Weiß and Swidan 2025). Classical Markov chain models have been widely used to capture transitions between AQI categories, but they suffer from exponential growth in the number of parameters as the state space increases. To mitigate this, extensions such as discrete ARMA (DARMA) processes (Jacobs and Lewis 1978a, 1978b, 1978c), mixture transition distribution (MTD) models (Raftery 1985; Berchtold and Raftery 2002), Pegram's operator-based ARMA process (Pegram 1980; Biswas and Song 2009) and regime-switching DARMA models (Weiß 2020) have been proposed. However, these approaches largely fail to address multiple seasonality, which are a defining feature of urban AQI series, nor do they flexibly incorporate exogenous drivers such as meteorological variables.

In parallel, machine learning methods, like random forests, support vector machines and deep learning, have shown strong predictive performance in air quality forecasting (e.g., Gupta, Kapadia, and Bhadane 2023; Mishra and Gupta 2024; Desai et al. 2025; Lim et al. 2021). Yet, these models are often criticized for limited interpretability and a lack of explicit representation of seasonal structures. By contrast, ordinal logistic regression (Agresti 2002; Fokianos and Kedem 2003) offers a statistically grounded and interpretable framework for modeling ordered categorical outcomes, while naturally accommodating external covariates such as temperature, humidity, traffic intensity and industrial activity. Its flexibility also allows for the incorporation of seasonal components, for instance through Fourier expansions or indicator functions. Despite this potential, the use of ordinal regression models to explicitly model multiple seasonality in categorical AQI series remains underexplored, particularly in urban contexts like Kolkata and Mumbai where both anthropogenic factors and climatic cycles strongly influence air quality dynamics. Here, we propose new ordinal logistic regression model incorporating Fourier series and indicator functions to model multiple seasonalities present in a categorical time series.

3 | Daily AQI Data of Kolkata

To motivate the proposed methodology with real-life applications, we consider two real-life datasets on daily Air Quality Index (AQI) from Kolkata and Mumbai cities. For the sake of brevity, we present the exploratory data analysis for Kolkata in

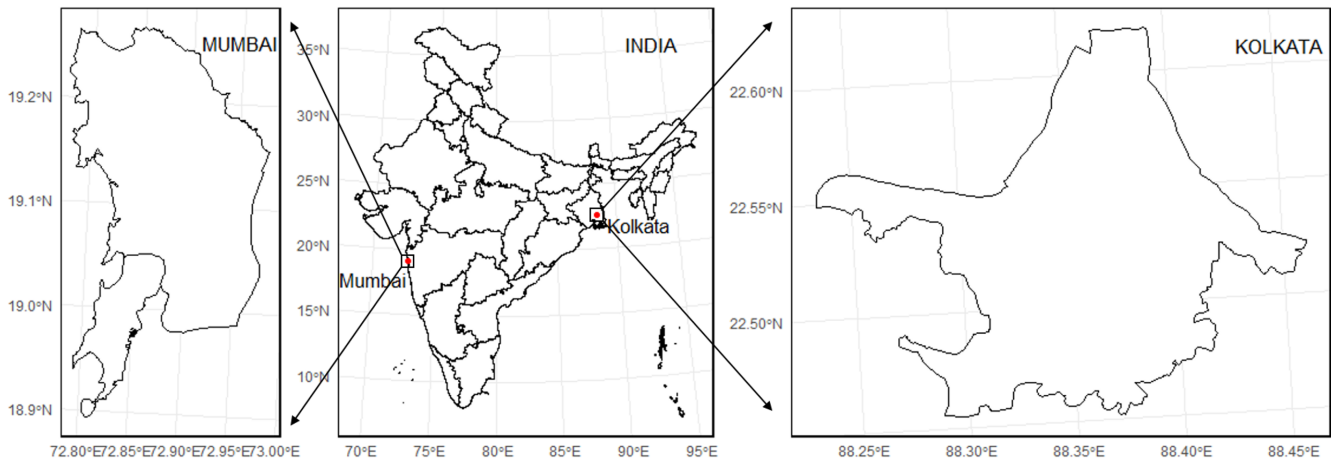


FIGURE 1 | Location map of Kolkata and Mumbai in India.

the main text, while a similar analysis for Mumbai is provided in the Appendix A.

The air quality index (AQI) is an indicator for reporting air quality daily. In India, the AQI is calculated using the concentrations of 12 types of air pollutants ascertained by Central Pollution Control Board (CPCB). For each pollutant, a sub-index, defined by CPCB, is calculated using its measured concentration. The overall AQI is the maximum of the individual sub-indices across all available pollutants. Detailed description of calculating AQI in India is discussed in “Understanding Air Pollution: A Guide Book” (2022).

3.1 | Study Area

This study focuses on the air quality data of Kolkata for the period 2019–2024. Kolkata is the capital city of the Indian state of West Bengal and it is the third most populous metropolitan region in India, following Delhi and Mumbai. It possesses a total area of 206.08 km² with a population of approximately 4.5 million and a population density of about 30,000 persons per km² (<https://en.wikipedia.org/wiki/Kolkata>). Geographically, the city is situated at 22°34′3″N and 88°22′12″E. The geographical position of Kolkata and Mumbai within India is illustrated in Figure 1. It serves as one of the major industrial and economic hubs of the country, contributing significantly to regional development. Among the twelve nationally monitored pollutants, only particulate matters (PM_{10} and $PM_{2.5}$) have been observed to exceed permissible limits during the winter season in Kolkata (see, e.g., “Understanding Air Pollution: A Guide Book” 2022).

The data of 2019–2024 is collected from official air quality monitoring stations operated by the Central Pollution Control Board (CPCB) and the West Bengal Pollution Control Board (WBPCB). These stations record daily pollutant concentrations. Data are publicly available through the AQI Data Repository of National Air Quality Index (NAQI) portal (<https://airquality.cpcb.gov.in/ccr/#/caaqm-dash-board-all/caaqm-landing/aqi-repository>), which provides a reliable basis for time series analysis and forecasting.

TABLE 1 | National AQI classification scheme.

Average AQI value over 24 h	Air quality category	Code
≤ 50	Good	0
51–100	Satisfactory	1
101–200	Moderate	2
201–300	Poor	3
301–400	Very poor	4
≥ 401	Severe	5

3.2 | Data Preprocessing and Preparation

For model development, the raw AQI data are first preprocessed to address missing or inconsistent entries. The final daily AQI values are then mapped to standard categorical labels following the National AQI classification scheme (<https://www.iitk.ac.in/new/national-air-quality-index>), as shown in Table 1. These categorical labels are subsequently encoded numerically in ordinal form to facilitate statistical modeling. The clean and ready-to-use daily AQI data of Kolkata from 2019 to 2024, along with the associated R and Python codes for data analysis, are available on the author’s GitHub repository: <https://github.com/ganirban004/aqi-logistic-paper.git>.

3.3 | Exploratory Analysis

Exploratory data analysis is performed to understand the temporal patterns and seasonality at different frequency. The original numerical AQI time series and the corresponding transformed categorical time series are presented in Figure 2a,b. Both representations exhibit a distinct seasonal pattern in air quality across the year. A clear decreasing trend is observed in the numerical AQI values, indicating gradual improvement in air quality. In the categorical series (Figure 2b), no instance of the “Severe” category is observed and notably, the last two winters show an absence of “Very Poor” days as well. These observations collectively suggest a substantial improvement in Kolkata’s air quality in recent years.

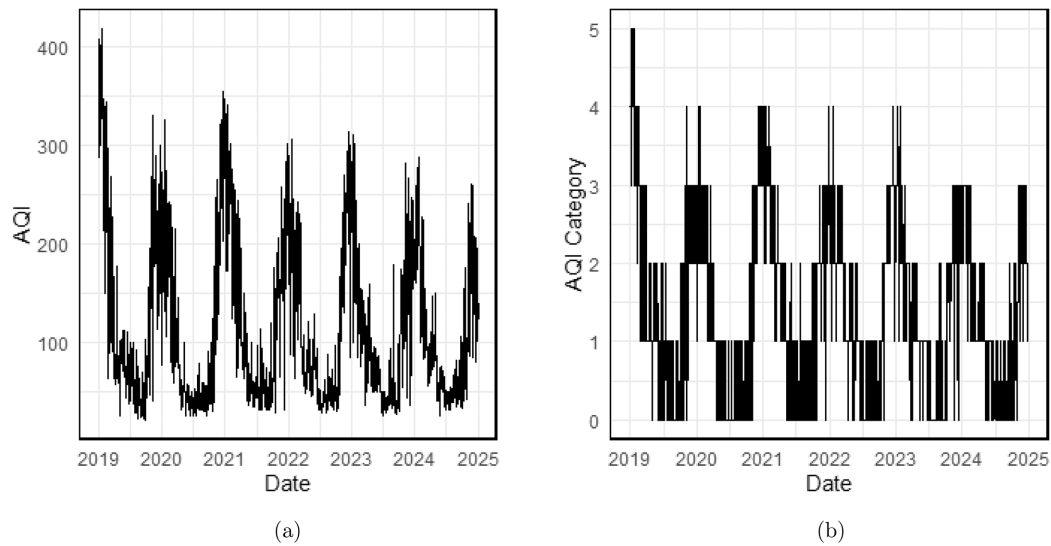


FIGURE 2 | Daily Air Quality Index (AQI) time series (original and categorical) of Kolkata during 2019–2024 showing the variation in air pollution levels over time with noticeable seasonal and episodic fluctuations. (a) Daily Air Quality Index (AQI) time series of Kolkata during 2019–2024 (higher values indicate poorer air quality), (b) Daily categorical AQI time series for Kolkata from 2019 to 2024 (higher category indicate poorer air quality).

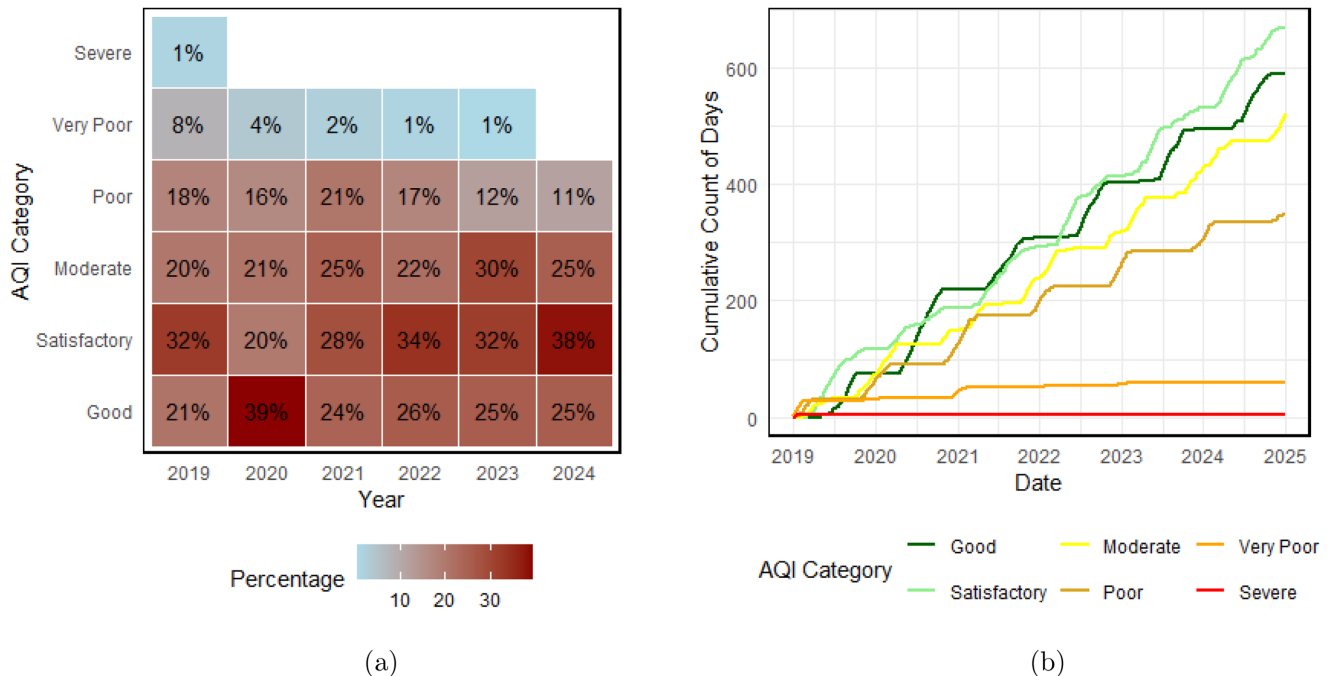


FIGURE 3 | Yearly trend and rate evolution graph of categorical AQI levels in Kolkata from 2019 to 2024. (a) Yearly Trend of categorical AQI levels in Kolkata from 2019 to 2024, (b) Daily rate evolution graph of categorical AQI levels in Kolkata from 2019 to 2024.

The temporal evolution of air quality in Kolkata is depicted in Figure 3a, showing the annual percentage distribution of days across different AQI categories from 2019 to 2024. A clear increasing trend is observed in the percentage of “Satisfactory” days, indicating progressive improvement in air quality over the study period, except for the year 2020. During 2020, corresponding to the COVID-19 lockdown, there is a sudden increase in the percentage of “Good” AQI days, due to reduced industrial and vehicular activity. In contrast, the shares of “Poor” and “Very Poor” categories show a steady decline and notably, no “Severe” AQI days are recorded after 2019 as seen in Figure 2b. Figure 3b shows

the evolution of rate of different AQI categories over time (see Weiß (2020) to know more about rate evolution graphs). Slope of this rate evolution graph gives the estimated marginal probability of the corresponding category. The curve for “Very Poor” and “Severe” categories are almost flattened in recent years.

The frequency plot shown in Figure 4a illustrates the prevalence of each air quality level throughout the study period (2019–2024). The transition probability matrix given in Figure 4b shows the probabilities of transitioning from one air quality category to another on consecutive days during the study period. Higher

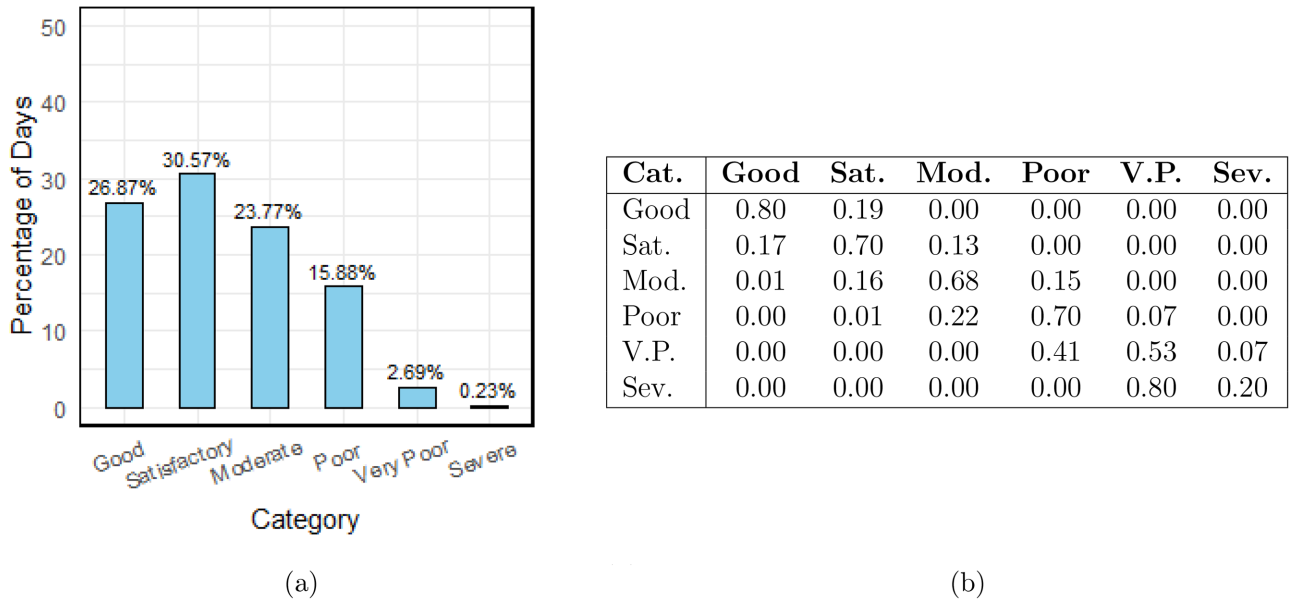


FIGURE 4 | Frequency distribution of AQI categories and Transition probability matrix for the categorical AQI time series of Kolkata. (a) Frequency distribution of AQI categories in Kolkata over the entire study period (2019–2024), (b) Transition Probability Matrix of Air Quality Categories over the entire study period (2019–2024).

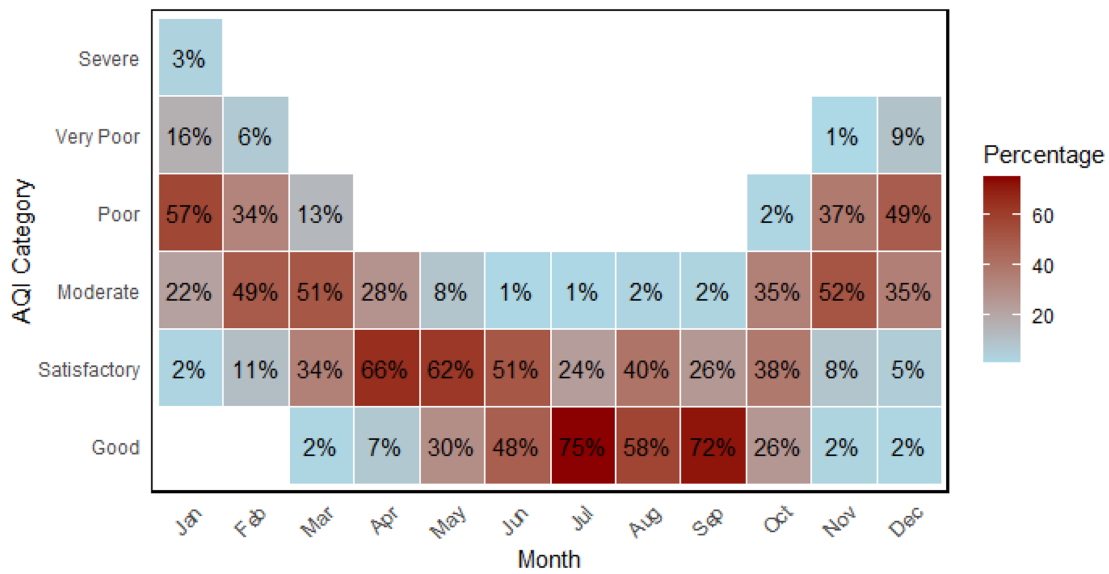


FIGURE 5 | Month-category heatmap of AQI categories using intensity measure for Kolkata.

values along the diagonal indicate strong temporal persistence within the same category, while nonzero off-diagonal entries reflect the likelihood of short-term changes in air quality conditions.

3.4 | Month-Category Heatmap Using Intensity Measure

To visualize the seasonal patterns, we construct a month-category heatmap using an intensity measure. Let $Y_t \in \{0, 1, 2, 3, 4, 5\}$ denote the AQI category on day t , encoded as Good = 0, Satisfactory = 1, Moderate = 2, Poor = 3, Very Poor = 4 and Severe = 5. Let $m \in \{1, \dots, 12\}$ denote the month of an observation. For month m and AQI category j , we define the frequency of occurrence as

$$f_{m,j} = \sum_{t \in \text{Month } m} \mathbb{I}(Y_t = j),$$

where $\mathbb{I}(\cdot)$ is the indicator function. Accordingly, we define the percentage of days in each category by dividing $f_{m,j}$ by n_m , total number of days in month m and then multiply by 100 as

$$p_{m,j} = \frac{f_{m,j}}{n_m} \times 100\%.$$

The heatmap is constructed by placing months along the horizontal axis and AQI categories along the vertical axis, with each cell's color intensity representing the corresponding percentage of days. Figure 5 illustrates this visualization. Darker shades denote higher percentages, offering an immediate visual

impression of the distribution of air quality levels across different months. This graphical representation effectively highlights seasonal patterns, such as the increased frequency of Poor and Very Poor air quality days during winter (December, January and February) and the predominance of Moderate or Good air quality days throughout the monsoon season (June, July, August and September).

3.5 | Multiple Seasonality of AQI Data

3.5.1 | Seasonal Patterns: Summer vs. Monsoon vs. Autumn vs. Winter

The plot, displayed in Figure 6, illustrates the variation in air quality across the four seasons namely summer (March, April and May), monsoon (June, July, August and September), autumn (October and November) and winter (December, January and February). Poorer air quality levels (e.g., “Moderate”,

“Poor” and “Very Poor”) are more prevalent during winter and post-monsoon seasons, while cleaner air conditions (e.g., “Good” and “Satisfactory”) are more common during the monsoon, highlighting the seasonal influence on pollution dynamics. The monsoon season typically exhibits better air quality, likely due to the cleansing effect of rainfall. These recurring seasonal fluctuations suggest the need to incorporate seasonal components or dummy variables when modeling time series.

3.5.2 | Seasonal Patterns: Weekdays vs. Weekends

The distribution of AQI categories across weekdays and weekends, depicted in Figure 7, exhibits subtle yet meaningful variations. On weekends, a higher proportion of days falls under the “Satisfactory” category (32.75%) compared to weekdays (29.69%), suggesting slightly better air conditions during weekends. Meanwhile, the proportion of “Good” days is marginally higher on weekdays (27.59%) than weekends (25.08%). The frequencies

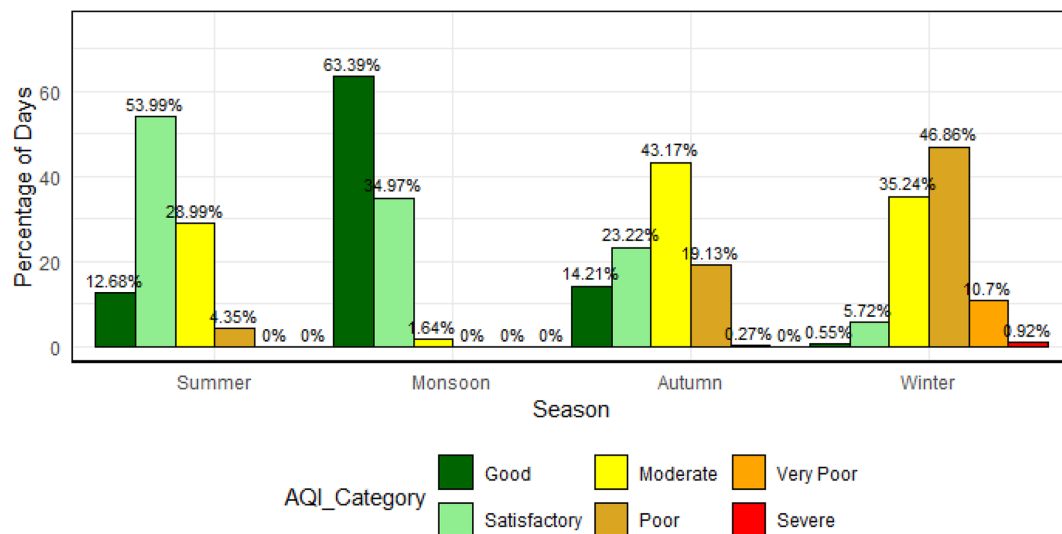


FIGURE 6 | Seasonal distribution of categorical AQI levels in Kolkata from 2019 to 2024.

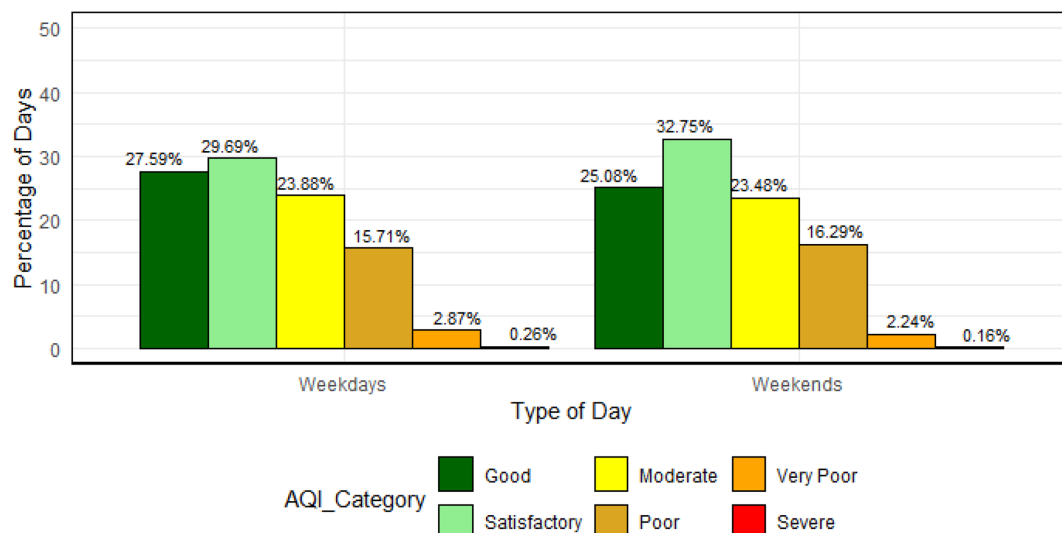


FIGURE 7 | Frequency distribution of AQI categories during weekdays and weekends in Kolkata (2019–2024).

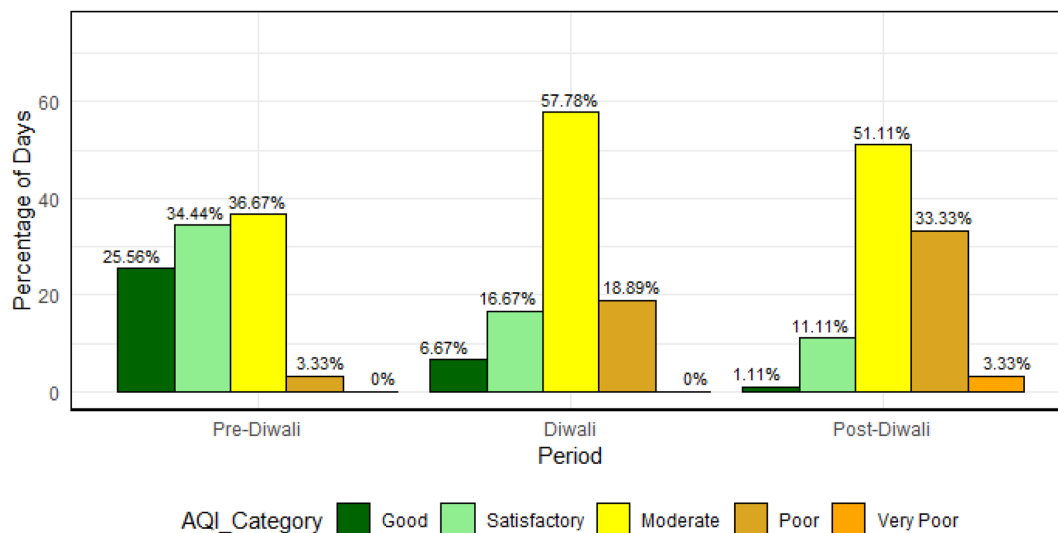


FIGURE 8 | Frequency distribution of AQI categories during the Pre-Diwali, Diwali and Post-Diwali periods in Kolkata (2019–2024).

of “Moderate” and “Poor” categories remain relatively consistent across both types of day, while the most hazardous categories, like “Very Poor” and “Severe”, occurred infrequently, with slightly higher occurrences on weekdays. These findings indicate that although air quality remains broadly similar throughout the week, weekends show a modest shift toward improved conditions, possibly due to decreased traffic and industrial activity.

3.5.3 | Seasonal Patterns: Festive vs. Non-Festive

Air pollution in Kolkata displays pronounced seasonal variability during festival like Diwali. To see the impact, the study period is divided into three sequential phases as follows:

- **Pre-Diwali Period (Day-22 to Day-8):** This period refers to the 15 days leading up to the Diwali window, specifically from 22 days before Diwali to 8 days before Diwali.

Pre-Diwali Period = [Diwali – 22 days, Diwali – 8 days]

- **Diwali Period (Day-7 to Day +7):** This is a 15-day symmetric window centered on the Diwali festival, including 7 days before and 7 days after Diwali.

Diwali Period = [Diwali – 7 days, Diwali + 7 days]

- **Post-Diwali Period (Day +8 to Day +22):** This period includes the 15 days following the Diwali window, from 8 days after Diwali to 22 days after Diwali.

Post-Diwali Period = [Diwali + 8 days, Diwali + 22 days]

This sequential structure allows for systematic analysis of air quality trends before, during and after the Diwali festival, isolating the impact of firecracker emissions and seasonal variations. The plot displayed in Figure 8 highlights a noticeable increase in the frequency of “Moderate”, “Poor” and “Very Poor” air quality days during the Diwali and Post-Diwali periods, indicating a significant short-term deterioration in air quality associated

with festival-related emissions. Figure 9 illustrates the yearly trend in AQI categories during the Diwali period. The percentage of days classified as “Good” increases from 7% in 2022 to 13% in both 2023 and 2024. Similarly, the percentages of both “Satisfactory” and “Moderate” categories rise to 40% in 2024, from 13% and 27%, respectively, in 2023. Moreover, the percentage of days falling under the “Poor” category declines substantially from 47% in 2023 to 7% in 2024. These observations suggest that the adverse impact of Diwali on air quality has weakened in recent years, possibly reflecting greater environmental awareness and increased health concern among the common people.

The descriptive figures suggest that the winter effect is stronger than the Diwali and weekend effects because the changes in AQI category proportions across seasons are much larger. The descriptive plots also indicate that the last two winters had no “Very Poor” AQI days, suggesting a long-term improvement in air quality during the study period (2019–2024). The Diwali and immediate post-Diwali periods show higher frequencies of poorer AQI categories compared with the pre-Diwali period. Moreover, the adverse effect of Diwali is getting weakened in recent years. Also, a small weekend effect is present. Weekends have a slightly higher proportion of Satisfactory days (32.75%) than weekdays (29.69%), along with a slightly lower proportion of the more polluted categories.

In summary, the exploratory data analysis offers clear evidence of multiple forms of seasonality in Kolkata’s AQI categories. Properly modeling these seasonal components is therefore essential for improving the accuracy of pollution assessment and supporting effective public-health and mitigation strategies.

4 | Proposed Model

In the context of electricity consumption data in Australia, (Liviera et al. 2011) proposed the TBATS model to capture multiple seasonality and trend. The TBATS model is designed to handle complex time series with multiple seasonality, nonlinear

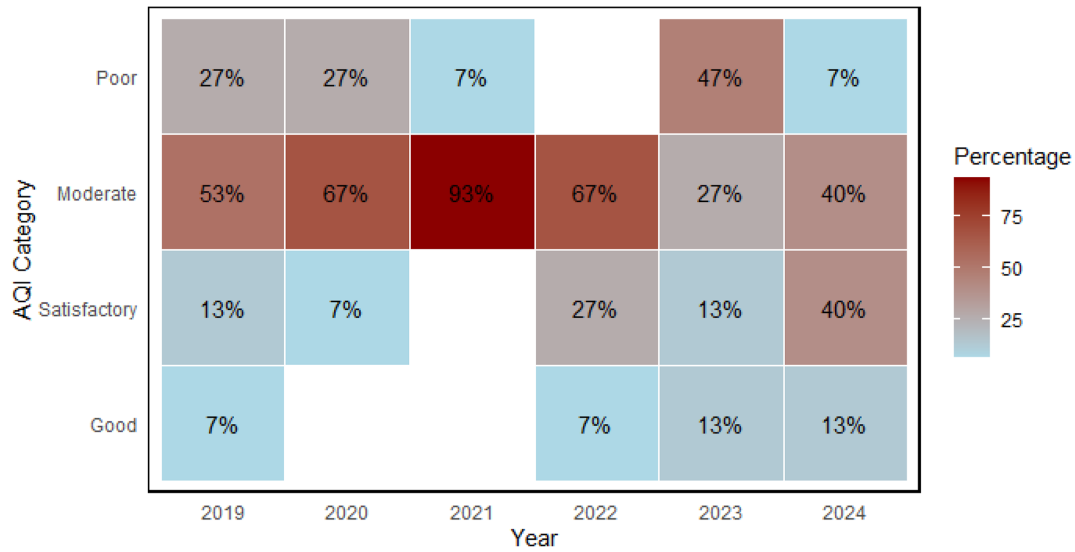


FIGURE 9 | Yearly trend of AQI categories during the Diwali period in Kolkata (2019–2024).

transformations and short-term dependencies in continuous time series. The acronym TBATS stands for Trigonometric seasonality, Box–Cox transformation, ARMA errors, Trend and Seasonal components. Let y_t denote the observed value of a time series at time point t . A Box–Cox transformation is first applied on the original time series y_t as

$$y_t^{(\omega)} = \begin{cases} \frac{y_t^\omega - 1}{\omega} & \text{for } \omega \neq 0 \\ \log y_t & \text{for } \omega = 0 \end{cases}$$

where ω is the transformation parameter. The transformed series is then modeled as

$$y_t^{(\omega)} = l_t + \phi b_t + \sum_{i=1}^T s_{i,t-P_i} + \varepsilon_t,$$

where l_t is the local level, b_t is the trend component with damping parameter $\phi \in (0, 1)$, $s_{i,t}$ are the seasonal components, T is the number of seasonal components and ε_t follows an ARMA(p, q) process. Each seasonal component is expressed using trigonometric terms based on a Fourier expansion. For a seasonal period P_i , the representation is

$$s_{i,t} = \sum_{k=1}^{K_i} \left[\beta_{1i}^{(k)} \cos\left(\frac{2\pi kt}{P_i}\right) + \beta_{2i}^{(k)} \sin\left(\frac{2\pi kt}{P_i}\right) \right]$$

where K_i is the number of harmonics chosen for seasonal cycle i . The state vectors $\beta_{1i}^{(k)}$ and $\beta_{2i}^{(k)}$ evolve dynamically according to recursive trigonometric updates. In its original formulation, the TBATS model effectively captures multiple seasonality, non-linear trends and short-term dependencies in continuous-valued time series. The use of Fourier terms, Box–Cox transformations and ARMA error structures allows TBATS to model complex periodic patterns across multiple timescales simultaneously. However, its direct applicability to categorical time series is limited because TBATS assumes numerical observations and additive errors, whereas categorical AQI data are non-numerical.

To bridge this gap, we propose an extension of the TBATS idea for ordinal categorical time series. Specifically, we employ a proportional odds model (Agresti 2002) for ordinal responses, with Fourier series terms as covariates to capture multiple seasonal patterns hidden in the AQI data. Accordingly, we refer to this framework as the Trigonometric Seasonality in Ordinal Logistic Regression (TSOLR) model. This approach preserves the key strength of TBATS, explicit modeling of multiple seasonal cycles, while respecting the discrete, ordered nature of AQI categories.

Let $Y_t \in \{0, 1, \dots, m\}$ denote the AQI category on day t . We assume that the underlying latent process is influenced by multiple seasonality, analogous to the TBATS framework, but adapted to an ordinal regression setting. We adopt the proportional odds model (Agresti 2002), where the cumulative probability of being in category j or below at time t is given by

$$\text{logit}(P(Y_t \leq j | \mathcal{F}_{t-1}, \mathbf{X}_t)) = \theta_j - \eta_t, \quad j = 0, 1, \dots, m-1 \quad (1)$$

where $\mathcal{F}_t$ contains all the past information up to time point t , that is, $\mathcal{F}_t = \{(Y_{t'}, \mathbf{X}_{t'}) : t' = 1, 2, \dots, t\}$, θ_j 's are category-specific threshold parameters and η_t 's are the linear predictor given in the form

$$\eta_t = \sum_{i=1}^T \sum_{k=1}^{K_i} [\beta_{1i}^{(k)} \cos(\frac{2\pi kt}{P_i}) + \beta_{2i}^{(k)} \sin(\frac{2\pi kt}{P_i})] + \mathbf{X}_t^\top \boldsymbol{\gamma},$$

where T is the number of seasonal components, P_i is the seasonal period for i^{th} seasonal component (e.g., in daily data, $P_i = 7$ for weekly seasonality, $P_i = 365$ for yearly seasonality etc.) and K_i is the number of Fourier harmonics chosen for a particular seasonal cycle i . $\beta_{1i}^{(k)}$ and $\beta_{2i}^{(k)}$ are Fourier coefficients for k^{th} harmonic corresponding to i^{th} seasonal component. The term $\mathbf{X}_t$ includes additional covariates, which may be purely exogenous (like humidity, temperature or particulate matter concentrations) or indicator variables corresponding to lagged AQI categories and $\boldsymbol{\gamma}$ be the corresponding parameter vector. In our data analysis, we consider the additional covariates as the indicator variables

corresponding to lag-1 AQI categories to capture the temporal dependencies. Then η_t depends on j and takes the form

$$\eta_{ij} = \sum_{i=1}^T \sum_{k=1}^{K_i} \left[\beta_{1i}^{(k)} \cos\left(\frac{2\pi kt}{P_i}\right) + \beta_{2i}^{(k)} \sin\left(\frac{2\pi kt}{P_i}\right) \right] + \sum_{k=0}^{m-1} \gamma_{jk} \mathbb{I}(Y_{t-1} = k)$$

The selection of the optimal K_i is carried out using a grid search procedure. Specifically, the model is fitted over a predefined set of candidate values for K_i and for each fitted model, the Akaike Information Criterion (AIC) is computed. The optimal values of K_i 's are then chosen as the one that minimizes the corresponding AIC value. In practice, the search space for each K_i is restricted to the discrete set $\{1, 2, \dots, \min(10, \lfloor \frac{P_i}{2} \rfloor)\}$. Empirical investigations indicate that this range is sufficient to capture the optimal choice of K_i for all i (for further details, see Appendix A). The final TSOLR model takes the form

$$\text{logit}(P(Y_t \leq j | \mathcal{F}_{t-1}, \mathbf{X}_t)) = \theta_j - \sum_{i=1}^T \sum_{k=1}^{K_i} \left[\beta_{1i}^{(k)} \cos\left(\frac{2\pi kt}{P_i}\right) + \beta_{2i}^{(k)} \sin\left(\frac{2\pi kt}{P_i}\right) \right] - \sum_{k=0}^{m-1} \gamma_{jk} \mathbb{I}(Y_{t-1} = k) \quad (2)$$

This formulation allows the ordinal AQI model to inherit the TBATS philosophy of explicitly modeling multiple seasonal components, while remaining suitable for ordered categorical outcomes. Inclusion of lagged category indicators captures the short-term dependencies in the categorical AQI.

In other classical approach, multiple seasonality are captured through indicator variables which are discrete functions as compared to Fourier series. We refer to this framework as the Indicator Seasonality in Ordinal Logistic Regression (ISOLR) model. For the daily AQI data, there are three seasonality: weekly-seasonality, seasonal-seasonality and festival-seasonality (Diwali vs Non-Diwali). Hence, the linear predictor η_t in model (1) takes the form

$$\eta_t = \beta_1 \cdot \text{Summer}_t + \beta_2 \cdot \text{Monsoon}_t + \beta_3 \cdot \text{Winter}_t + \beta_4 \cdot \text{Diwali}_t + \beta_5 \cdot \text{Weekend}_t + \mathbf{X}_t^\top \boldsymbol{\gamma}$$

where Summer_t , Monsoon_t , and Winter_t are binary variables, which take the value 1 if day t falls in respective season and 0 otherwise. The autumn season is considered as the baseline category. Similarly, Diwali_t takes the value 1 if day t falls in the 15-day Diwali period and Weekend_t takes the value 1 if day t is either Saturday or Sunday, 0 otherwise. $\mathbf{X}_t$ is additional covariate, which may be purely exogenous or indicator variables corresponding to lagged AQI categories and $\boldsymbol{\gamma}$ is the corresponding parameter vector. Considering the additional covariate as the lag-1 AQI categories, the final ISOLR model takes the form

$$\log\left(\frac{P(Y_t \leq j | \mathcal{F}_{t-1}, \mathbf{X}_t)}{P(Y_t > j | \mathcal{F}_{t-1}, \mathbf{X}_t)}\right) = \theta_j - (\beta_1 \cdot \text{Summer}_t + \beta_2 \cdot \text{Monsoon}_t + \beta_3 \cdot \text{Winter}_t + \beta_4 \cdot \text{Diwali}_t + \beta_5 \cdot \text{Weekend}_t) - \sum_{k=0}^{m-1} \gamma_{jk} \mathbb{I}(Y_{t-1} = k) \quad (3)$$

4.1 | Interpretation of Parameters

The interpretation of the Fourier coefficients is carried out jointly rather than individually. For a given seasonal component, the term

$$\beta_1^{(k)} \cos\left(\frac{2\pi kt}{P}\right) + \beta_2^{(k)} \sin\left(\frac{2\pi kt}{P}\right)$$

can be equivalently written as

$$A_k \cos\left(\frac{2\pi kt}{P} - \phi_k\right),$$

where $A_k = \sqrt{(\beta_1^{(k)})^2 + (\beta_2^{(k)})^2}$ represents the amplitude and $\phi_k = \tan^{-1} \frac{\beta_2^{(k)}}{\beta_1^{(k)}}$ represents the phase shift. Hence, the Fourier coefficients collectively determine the strength and timing of seasonal variation in AQI. Larger amplitudes indicate stronger seasonal fluctuations, while the phase determines when peaks and troughs occur.

In the ISOLR model, seasonality and festive effects are captured using indicator variables, leading to a more direct interpretation. Each coefficient β_i associated with an indicator variable (e.g., Summer, Monsoon, Winter, Diwali, Weekend) represents the change in the cumulative probabilities of the AQI categories when the indicator is active (equal to 1), relative to the baseline category (when the indicator is 0). Specifically, a positive coefficient implies that the corresponding season or event increases the likelihood of higher AQI categories, while a negative coefficient indicates an improvement in air quality relative to the baseline. This interpretation can also be expressed in terms of odds ratio. For a seasonal indicator variable with coefficient β_i , the quantity $\exp(-\beta_i)$ represents the multiplicative change in the cumulative odds when the indicator variable changes value from 0 to 1, holding all other variables constant. For example, if $\exp(-\beta_i) < 1$, the odds of being in lower AQI categories decrease, implying a shift toward higher (worse) categories. If $\exp(-\beta_i) > 1$, the odds of being in lower categories increase, indicating an improvement in AQI.

5 | Estimation of Parameters

To estimate the parameters of the proposed models, we maximize the log-likelihood function. To construct the likelihood function, we first write the model (1) in the form of a linear model as follows.

$$\text{logit}(P(Y_t \leq j | \mathcal{F}_{t-1}, \mathbf{X}_t)) = \boldsymbol{\psi}_{1 \times p}^\top \mathbf{D}_t^{p \times 1}, \quad t = 1, 2, \dots, N$$

where $\boldsymbol{\psi}_{p \times 1} = (\boldsymbol{\theta}^\top, \boldsymbol{\beta}^\top, \boldsymbol{\gamma}^\top)^\top$ be the full parameter vector and $\mathbf{D}_t^{p \times 1}$ includes the values of all the covariates at time t corresponding to model (1). then the log-likelihood is defined as

$$\begin{aligned} \ell(\boldsymbol{\psi}) &= \log \prod_{t=1}^N \prod_{j=0}^m P(Y_t = j | \mathcal{F}_{t-1}, \mathbf{X}_t)^{\mathbb{I}(Y_t=j)} \\ &= \sum_{t=1}^N \sum_{j=0}^m \mathbb{I}(Y_t = j) \log P(Y_t = j | \mathcal{F}_{t-1}, \mathbf{X}_t) \end{aligned} \quad (4)$$

where

$$P(Y_t = j | \mathcal{F}_{t-1}, \mathbf{X}_t) = \begin{cases} P(Y_t \leq j | \mathcal{F}_{t-1}, \mathbf{X}_t) & \text{for } j = 0 \\ P(Y_t \leq j | \mathcal{F}_{t-1}, \mathbf{X}_t) - P(Y_t \leq j-1 | \mathcal{F}_{t-1}, \mathbf{X}_t) & \text{for } j = 1, 2, \dots, m-1 \\ 1 - P(Y_t \leq m-1 | \mathcal{F}_{t-1}, \mathbf{X}_t) & \text{for } j = m \end{cases}$$

and

$$P(Y_t \leq j | \mathcal{F}_{t-1}, \mathbf{X}_t) = \begin{cases} \frac{\exp(\theta_j - \eta_t)}{1 + \sum_{i=0}^{m-1} \exp(\theta_i - \eta_t)} & \text{if } j = 0, 1, \dots, m-1 \\ \frac{1}{1 + \sum_{i=0}^{m-1} \exp(\theta_i - \eta_t)} & \text{if } j = m \end{cases}$$

as described in model (1). Then the log-likelihood (4) is maximized with respect to the parameter vector $\boldsymbol{\psi}$ to get the maximum likelihood estimate $\hat{\boldsymbol{\psi}} = (\hat{\boldsymbol{\theta}}^\top, \hat{\boldsymbol{\beta}}^\top, \hat{\boldsymbol{\gamma}}^\top)^\top$. Under some regularity conditions, it is well established fact that the MLEs are asymptotically normal which is given by

$$\sqrt{N}(\hat{\boldsymbol{\psi}} - \boldsymbol{\psi}) \xrightarrow{d} \mathcal{N}(0, \mathbf{I}^{-1}(\boldsymbol{\psi})).$$

The observed Fisher information matrix $\mathbf{I}(\boldsymbol{\psi})$ is computed as the negative of the Hessian matrix of the log-likelihood at the MLE:

$$\hat{\mathbf{I}}(\boldsymbol{\psi}) = -\nabla^2 \ell(\boldsymbol{\psi}) \Big|_{\boldsymbol{\psi}=\hat{\boldsymbol{\psi}}}$$

This Hessian matrix is computed numerically and it serves as the variance-covariance matrix of the estimated parameters, which can be further used to construct confidence interval of the estimated parameters. For parameter $\boldsymbol{\psi}$, the partial derivatives are approximated using finite differences of the log-likelihood function $\ell(\boldsymbol{\psi})$. The step size for calculating finite differences is chosen to balance numerical precision and stability.

6 | Existing Models

In order to compare the forecasting performance of TSOLR and ISOLR models, we list some existing methodologies for modeling categorical time series in this section. The list includes some Markov chain based models along with some machine learning and deep learning models.

6.1 | Markov Chain Model

A traditional approach for modeling categorical time series is the use of Markov Chains of order p , where the transition probability is defined as:

$$P(Y_t = j | Y_{t-1}, \dots, Y_{t-p}), \quad j = 0, 1, \dots, m.$$

However, this approach suffers from an exponential increase in the number of parameters as the order increases. For a Markov chain of order p with $(m+1)$ categories, the number of

parameters is $m(m+1)^p$. For example, with $m = 4$ and $p = 2$, 100 parameters are required, which is computationally intensive. Furthermore, this model cannot incorporate external covariates.

6.2 | Mixture Transition Distribution (MTD) Model

To reduce complexity, the MTD model was introduced by Raftery (1985) and later refined by Berchtold and Raftery (2002). It models transition probabilities as a weighted mixture of one-step transitions:

$$P(Y_t | Y_{t-1}, \dots, Y_{t-p}) = \sum_{k=1}^p \lambda_k P(Y_t | Y_{t-k}),$$

where $\lambda_k \geq 0$ and $\sum \lambda_k = 1$. The MTD model reduces the parameter count to $m(m+1) + p - 1$. For example, with $m = 4$ and $p = 2$, the parameter count drops to 21. Like the Markov chain model, this approach also does not consider information of any external covariate.

6.3 | Pegram's Autoregressive (PAR) Process

Pegram (1980) proposed a simplified approach called the PAR(p) process, a special case of the discrete ARMA model (Jacobs and Lewis 1978a, 1978b, 1978c). The PAR process is defined by:

$$P(Y_t = j | Y_{t-1}, \dots, Y_{t-p}) = \sum_{k=1}^p \phi_k \mathbb{I}(Y_{t-k} = j) + \left(1 - \sum_{k=1}^p \phi_k\right) \pi_j,$$

where $\phi_k \in (0, 1)$ and $\sum_{k=1}^p \phi_k < 1$, π_j is the marginal probability of category j and $\mathbb{I}(\cdot)$ is the indicator function. However, this model does not include external covariate in the model.

6.4 | Decision Trees and Random Forests (RF) for Ordinal Data

Standard tree-based methods can be adapted conveniently (Janitza et al. 2016) using ordinal splitting criteria like the quadratic entropy index or the use of monotonic constraints. Random Forest (RF) is an ensemble-based learning method that extends traditional decision tree models by combining multiple trees trained on random subsets of the data and features. Let $\mathcal{D} = \{(\mathbf{X}_t, \mathbf{Z}_t, Y_t)\}_{t=1}^N$ denote the training dataset, where $\mathbf{X}_t$ and $\mathbf{Z}_t$ represent the additional covariates and seasonal indicators respectively. For each of the K bootstrap samples $\mathcal{D}_b^{(k)}$, a classification tree $h_k(X)$ is fitted using a random subset of predictors at each split to reduce correlation among trees. The final Random Forest prediction is obtained through majority voting:

$$\hat{y} = \arg \max_{j \in C} \sum_{k=1}^K \mathbb{I}(h_k(X) = j),$$

where $C = 0, 1, 2, \dots, m$ is the set of AQI categories and $\mathbb{I}(\cdot)$ is the indicator function.

Random Forests are known for their robustness against overfitting, ability to model nonlinear and complex relationships and strong predictive performance even with high-dimensional categorical features. However, their ensemble nature reduces interpretability and increases computational cost, posing challenges for theoretical inference and causal analysis in time-dependent categorical data.

6.5 | Support Vector Machines (SVMs)

Support Vector Machines (SVMs) are powerful supervised learning algorithms widely applied to classification problems (Awad and Khanna 2015). For a categorical time series $\{Y_t\}$, the predictor vector at time t may include both seasonal indicators and additional covariates. Given training data $D = \{(X_t, Z_t, Y_t)\}_{t=1}^N$ with $Y_t \in \{-1, +1\}$, a standard SVM seeks a separating hyperplane $w_1^\top X + w_2^\top Z + b = 0$ by solving

$$\min_{w_1, w_2, b, \xi} \frac{1}{2} (\|w_1\|^2 + \|w_2\|^2) + C \sum_{i=1}^N \xi_i, \quad \text{subject to}$$

$$Y_i(w_1^\top X + w_2^\top Z + b) \geq 1 - \xi_i, \quad \xi_i \geq 0,$$

where $C > 0$ regulates the balance between margin maximization and misclassification. For ordinal-valued Y_t (e.g., AQI categories), ordinal SVMs generalize this framework by introducing a sequence of ordered parallel hyperplanes,

$$\begin{aligned} w_1^\top X_t + w_2^\top Z_t - b_1 &= 0, \\ w_1^\top X_t + w_2^\top Z_t - b_2 &= 0, \quad \dots, \\ w_1^\top X_t + w_2^\top Z_t - b_{m-1} &= 0, \end{aligned}$$

ensuring that predicted classes respect their natural ordering.

SVMs demonstrate strong generalization in high-dimensional feature spaces and are robust against overfitting. However, their performance depends critically on the choice of kernel function and regularization parameters. Moreover, computational complexity of SVM increases with sample size.

6.6 | Recurrent Neural Networks (RNNs)

Recurrent Neural Networks (RNNs) and their advanced variants, such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) architectures, are widely used for modeling temporal dependencies in sequential data (Lindemann et al. 2021). Unlike traditional feedforward networks, RNNs incorporate feedback connections that allow information to persist across time steps. In this setting, the predictor sequence at time t may include both lagged category indicators as integers and exogenous variables, i.e., $X_t = (Y_{t-1}, \dots, Y_{t-p}, Z_{1t}, \dots, Z_{qt})^\top$. Given an input sequence $\{X_t\}_{t=1}^N$, a standard RNN updates its hidden state according to

$$h_t = \sigma(W_h h_{t-1} + W_x X_t + b_h),$$

where h_t denotes the hidden representation at time t , $\sigma(\cdot)$ is a nonlinear activation function and W_h, W_x, b_h are the model parameters. LSTMs enhance this framework by introducing gating mechanisms that regulate information flow and mitigate the vanishing

gradient problem. Specifically, for each time step t , the LSTM computes

$$\begin{aligned} f_t &= \sigma(W_f X_t + U_f h_{t-1} + b_f), \\ i_t &= \sigma(W_i X_t + U_i h_{t-1} + b_i), \\ \tilde{c}_t &= \tanh(W_c X_t + U_c h_{t-1} + b_c), \\ c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t, \\ o_t &= \sigma(W_o X_t + U_o h_{t-1} + b_o), \\ h_t &= o_t \odot \tanh(c_t), \end{aligned}$$

where f_t , i_t and o_t denote the forget, input and output gates respectively, $\odot$ represents element-wise multiplication and c_t is the memory cell state. Like decision tree and SVM, LSTM is also fitted using both types of covariates, namely, indicator variables and trigonometric terms as seasonal components.

For ordinal categorical outputs, a softmax output layer can be replaced or modified with an ordinal-aware loss, such as ordinal cross-entropy or cumulative link-based objectives, to preserve the intrinsic order among categories. Such architectures effectively capture nonlinear dynamics and long-range temporal dependencies, making them highly suitable for forecasting AQI categories over time. However, these models require extensive hyperparameter tuning to balance model complexity and generalization. Training can be computationally intensive due to sequential dependencies and like other deep learning methods, interpretability remains a challenge despite their strong predictive capability.

6.7 | Temporal Fusion Transformers (TFT)

The Temporal Fusion Transformers (Lim et al. 2021) is a modern deep learning architecture designed for interpretable multi-horizon time series forecasting. It combines recurrent neural networks with attention mechanisms to capture both short-term temporal dependencies and long-term relationships in sequential data. Given a sequence of inputs $\{X_t\}_{t=1}^T$, which may include historical observations, known future inputs and static covariates, TFT employs gated residual networks (GRNs) to select relevant features and suppress noise. The temporal dynamics are modeled using sequence-to-sequence structures, typically based on LSTM layers, while multi-head attention mechanisms are used to identify important time steps and dependencies, making the model particularly well-equipped to handle complex seasonal patterns.

A key component of TFT is the variable selection network, which adaptively assigns importance weights to input variables at each time step. The model outputs forecasts $\hat{Y}_{t+h}$ for multiple horizons $h = 1, \dots, H$, often optimized using quantile loss to capture predictive uncertainty. Formally, the model can be expressed as

$$\hat{Y}_{t+h} = f_\theta(X_{1:t}, X_{t+1:t+h}, S),$$

where S denotes static features and $f_\theta(\cdot)$ represents the TFT architecture parameterized by θ .

TABLE 2 | Various auto-association measures for categorical time series.

Measure	Definition	Range
Cohen's κ	$\kappa(h) = \frac{\sum_{i=0}^m p_{ii}^{(h)} - \sum_{i=0}^m p_{i\cdot}^{(h)} p_{\cdot i}^{(h)}}{1 - \sum_{i=0}^m p_{i\cdot}^{(h)} p_{\cdot i}^{(h)}}$	$\left[-\frac{\sum_{i=0}^m p_{i\cdot}^{(h)} p_{\cdot i}^{(h)}}{1 - \sum_{i=0}^m p_{i\cdot}^{(h)} p_{\cdot i}^{(h)}}, 1\right]$
Cramer's ν	$\nu(h) = \sqrt{\frac{1}{nm} \sum_{i=0}^m \sum_{j=0}^m \frac{(N_{ij}^{(h)} - E_{ij})^2}{E_{ij}}}$	$[0, 1]$
Goodman-Krushkal's τ	$A_{\nu}^{(\tau)}(h) = \frac{\sum_{i,j=0}^m \frac{p_{ij}^{(h)} p_{ji}^{(h)}}{p_{i\cdot}^{(h)} p_{\cdot j}^{(h)}} - \sum_{i=0}^m p_{i\cdot}^{(h)} p_{\cdot i}^{(h)}}{1 - \sum_{i=0}^m p_{i\cdot}^{(h)} p_{\cdot i}^{(h)}}$	$[0, 1]$
Goodman-Krushkal's γ	$\gamma = \frac{\text{no. of concordant pairs} - \text{no. of discordant pairs}}{\text{no. of concordant pairs} + \text{no. of discordant pairs}}$	$[-1, 1]$
Pearson Measure X_n^2	$X_n^2 = n \sum_{i=0}^m \sum_{j=0}^m \frac{(p_{ij}^{(h)} - p_{i\cdot}^{(h)} p_{\cdot j}^{(h)})^2}{p_{i\cdot}^{(h)} p_{\cdot j}^{(h)}}$	$[0, n(m-1)]$
Mutual Information	$A_{\nu}^{(u)}(h) = -\frac{\sum_{i,j=0}^m p_{ij}^{(h)} \ln\left(\frac{p_{ij}^{(h)}}{p_{i\cdot}^{(h)} p_{\cdot j}^{(h)}}\right)}{\sum_{i=0}^m p_{i\cdot}^{(h)} \ln p_{i\cdot}^{(h)}}$	$[0, 1]$

The TFT framework is particularly suitable for complex time series with nonlinear patterns and multiple covariates, offering both high predictive performance and interpretability through attention weights and feature importance scores. However, the model is computationally intensive and requires careful tuning of hyperparameters for stable training.

These machine learning methods offer flexible frameworks for modeling ordinal categorical time series and can be chosen based on data characteristics, interpretability requirements and computational considerations. In simulation study (Section 8) and data analysis (Section 9), we compare the performance of our proposed models with these benchmark methods.

7 | Auto-Dependency Measures for Categorical Time Series

Autocorrelation function (ACF) is a standard tool for measuring serial dependence in numeric time series. ACF plots are also used rigorously to determine the suitable order of a model. However, it is not directly applicable to categorical data due to the absence of a natural arithmetic structure. To study dependence in categorical time series, especially when data are nominal or ordinal, several ACF-like measures have been proposed.

Let $Y_t \in \{0, 1, \dots, m\}$ be a categorical time series. Define the joint probability at lag h as $p_{ij}^{(h)} = P(Y_t = i, Y_{t-h} = j)$, the marginal probabilities as $p_{i\cdot}^{(h)} = P(Y_{t-h} = i) = \sum_{j=0}^m p_{ij}^{(h)}$ and $p_{\cdot j}^{(h)} = P(Y_t = j) = \sum_{i=0}^m p_{ij}^{(h)}$. Under this setup, (Weiß and Göb 2008) summarized various auto-association measures, which are provided in Table 2. In the present analysis, these measures are used in the data analysis to select the appropriate model order.

8 | Simulation Study

This section presents a comprehensive simulation study to evaluate the performance of the proposed TSOLR and ISOLR models. The study consists of three parts. First, the empirical consistency of the maximum likelihood estimators is examined using data generated from the proposed models under different parameter settings and sample sizes. Second, the forecasting performance of the proposed models is compared with several existing

methods using data generated from the ISOLR model. Finally, the robustness of the proposed methods is investigated under alternative data-generating mechanisms, including a cumulative ordered probit model and a continuous latent variable model.

8.1 | Empirical Consistency

8.1.1 | TSOLR Model

To evaluate the empirical consistency of the TSOLR model, data are generated from the following TSOLR model with three seasonal components having seasonal frequencies of 100, 25, and 5, ensuring the presence of multiple seasonality in the data generating process.

$$\log\left(\frac{P(Y_t \leq j)}{P(Y_t > j)}\right) = \theta_j - \sum_{i=1}^T [\beta_{1i} \cdot \sin\left(\frac{2\pi t}{P_i}\right) + \beta_{2i} \cdot \cos\left(\frac{2\pi t}{P_i}\right)] - \sum_{k=0}^{m-1} \gamma_{jk} \mathbb{I}(Y_{t-1} = k)$$

where $j = 0, 1, \dots, m-1$, $T = 3$ is the number of seasonal components, $P_1 = 100$, $P_2 = 25$ and $P_3 = 5$. Two different sets of parameters are considered. The first setting corresponds to a three-category response, while the second corresponds to a four-category response. The parameter values used for both cases are given below.

- **Three-category case:** $m = 2$ with parameters $(\theta_0, \theta_1, \beta_{11}, \beta_{21}, \beta_{12}, \beta_{22}, \beta_{13}, \beta_{23}) = (-0.84, 0.84, 0, 2, 0, 0.7, 0, 0.2)$ with $\Gamma = ((\gamma_{jk}))_{2 \times 2} = \begin{bmatrix} -1.0 & 0.5 \\ -0.8 & 0.3 \end{bmatrix}$ and
- **Four-category case:** $m = 3$ with parameters $(\theta_0, \theta_1, \theta_2, \beta_{11}, \beta_{21}, \beta_{12}, \beta_{22}, \beta_{13}, \beta_{23}) = (-1.1, 0, 1.9, 0.6, -1.1, 1.4, -0.8, -0.1, -0.4)$ with $\Gamma = ((\gamma_{jk}))_{3 \times 3} = \begin{bmatrix} -1.2 & -0.5 & 0.3 \\ -0.8 & -0.3 & 0.2 \\ -0.4 & -0.1 & 0.1 \end{bmatrix}$.

For each parameter setting, datasets of sizes 500, 1000, and 10,000 are generated. For each sample size, the corresponding true model is fitted using maximum likelihood estimation. This process is repeated 1000 times using Monte Carlo simulation. The mean and standard deviation of the MLEs obtained from these

TABLE 3 | Average estimated values and standard errors (in parenthesis) of parameters when data generated from TSOLR model for three-category case.

Parameters	$n = 500$	$n = 1000$	$n = 10,000$
$\theta_0 = -0.84$	-0.81 (0.24)	-0.82 (0.17)	-0.84 (0.05)
$\theta_1 = 0.84$	0.91 (0.20)	0.88 (0.14)	0.84 (0.04)
$\beta_{11} = 0$	0.00 (0.13)	0.00 (0.09)	0.00 (0.03)
$\beta_{21} = 2$	2.07 (0.22)	2.04 (0.14)	2.00 (0.04)
$\beta_{12} = 0$	0.00 (0.14)	0.00 (0.10)	0.00 (0.03)
$\beta_{22} = 0.7$	0.72 (0.16)	0.71 (0.11)	0.70 (0.03)
$\beta_{13} = 0$	0.00 (0.14)	0.00 (0.10)	0.00 (0.03)
$\beta_{23} = 0.2$	0.21 (0.14)	0.20 (0.10)	0.20 (0.03)
$\gamma_{00} = -1.0$	-0.93 (0.33)	-0.96 (0.22)	-1.00 (0.07)
$\gamma_{01} = 0.5$	0.57 (0.38)	0.53 (0.26)	0.50 (0.08)
$\gamma_{10} = -0.8$	-0.74 (0.34)	-0.76 (0.22)	-0.80 (0.07)
$\gamma_{11} = 0.3$	0.36 (0.28)	0.33 (0.19)	0.30 (0.06)

TABLE 4 | Average estimated values and standard errors (in parenthesis) of parameters when data generated from TSOLR model for four-category case.

Parameters	$n = 500$	$n = 1000$	$n = 10,000$
$\theta_0 = -1.1$	-1.10 (0.35)	-1.09 (0.25)	-1.10 (0.07)
$\theta_1 = 0$	0.05 (0.28)	0.03 (0.20)	0.00 (0.06)
$\theta_2 = 1.9$	2.00 (0.28)	1.95 (0.19)	1.91 (0.06)
$\beta_{11} = 0.6$	0.62 (0.14)	0.61 (0.10)	0.60 (0.03)
$\beta_{21} = -1.1$	-1.14 (0.16)	-1.12 (0.11)	-1.10 (0.04)
$\beta_{12} = 1.4$	1.44 (0.16)	1.42 (0.11)	1.40 (0.03)
$\beta_{22} = -0.8$	-0.83 (0.15)	-0.82 (0.10)	-0.80 (0.03)
$\beta_{13} = -0.1$	-0.11 (0.14)	-0.11 (0.09)	-0.10 (0.03)
$\beta_{23} = -0.4$	-0.40 (0.14)	-0.40 (0.09)	-0.40 (0.03)
$\gamma_{00} = -1.2$	-1.17 (0.40)	-1.18 (0.29)	-1.20 (0.09)
$\gamma_{01} = -0.5$	-0.49 (0.45)	-0.49 (0.33)	-0.50 (0.10)
$\gamma_{02} = 0.3$	0.31 (0.43)	0.30 (0.31)	0.31 (0.09)
$\gamma_{10} = -0.8$	-0.74 (0.35)	-0.76 (0.25)	-0.80 (0.08)
$\gamma_{11} = -0.3$	-0.26 (0.39)	-0.27 (0.27)	-0.29 (0.08)
$\gamma_{12} = 0.2$	0.23 (0.34)	0.21 (0.24)	0.21 (0.07)
$\gamma_{20} = -0.4$	-0.33 (0.44)	-0.37 (0.30)	-0.39 (0.09)
$\gamma_{21} = -0.1$	-0.05 (0.45)	-0.07 (0.32)	-0.10 (0.09)
$\gamma_{22} = 0.1$	0.15 (0.32)	0.12 (0.23)	0.10 (0.07)

1000 replications are reported in Tables 3 and 4. The results show that the mean estimates approach the true parameter values as the sample size increases, while the standard errors decrease. These findings provide empirical evidence for the consistency of the proposed TSOLR model under different parameter settings.

8.1.2 | ISOLR Model

To evaluate the empirical consistency of the ISOLR model, data are generated from the following ISOLR model with different parameter settings.

$$\log\left(\frac{P(Y_t \leq j)}{P(Y_t > j)}\right) = \theta_j - \left(\beta_1 S_{1t}^{(1)} + \beta_2 S_{1t}^{(2)} + \beta_3 S_{2t} + \beta_4 S_{3t}\right) - \sum_{k=0}^{m-1} \gamma_{jk} \mathbb{I}(Y_{t-1} = k),$$

where $j = 0, 1, \dots, m-1$ and θ_j denotes the threshold parameter. Whereas $S_{1t}^{(1)}$ and $S_{1t}^{(2)}$ represent the annual seasonal effects, S_{2t} represents the weekly seasonal effect, S_{3t} represents the festive-period effect to mimic the real AQI data and γ_{jk} captures the first-order dependence through the previous response category.

The annual seasonal effect is represented using two binary indicator variables $S_{1t}^{(1)}$ and $S_{1t}^{(2)}$ as we assume three seasons in the data generating process. Here 100-day period is treated as one year and is divided into three seasons. The first season consists of days 1–30, the second season consists of days 31–70 and the remaining days 71–100 form the third season, which serves as the reference season. The first two seasons are represented by the binary variables

$$S_{1t}^{(1)} = \begin{cases} 1, & \text{if } t \in [100k + 1, 100k + 30], \\ 0, & \text{otherwise,} \end{cases} \quad k = 0, 1, \dots, \frac{n}{100} - 1,$$

and

$$S_{1t}^{(2)} = \begin{cases} 1, & \text{if } t \in [100k + 31, 100k + 70], \\ 0, & \text{otherwise,} \end{cases} \quad k = 0, 1, \dots, \frac{n}{100} - 1.$$

Thus, when both $S_{1t}^{(1)}$ and $S_{1t}^{(2)}$ are equal to zero, the observation belongs to the third season (days 71–100). Together, these two indicator variables completely describe the annual seasonal pattern.

The weekly seasonal effect is represented by the binary variable

$$S_{2t} = \begin{cases} 1, & \text{if } t \in [5k + 4, 5k + 5], \\ 0, & \text{otherwise,} \end{cases} \quad k = 0, 1, \dots, \frac{n}{5} - 1.$$

Here, each 5-day period is treated as one week and the last two days are regarded as weekends.

The festive-period seasonal effect is represented by the binary variable

$$S_{3t} = \begin{cases} 1, & \text{if } t \in [100k + 81, 100k + 85], \\ 0, & \text{otherwise,} \end{cases} \quad k = 0, 1, \dots, \frac{n}{100} - 1.$$

This variable represents a five-day festive period within each simulated year. As with the TSOLR model, here also two parameter settings with same sample sizes are considered: one for the three-category case and another for the four-category case.

- **Three-category case:** $m = 2$ with parameters $(\theta_0, \theta_1, \beta_1, \beta_2, \beta_3, \beta_4) = (-0.78, 0.75, -0.3, 1.2, -0.4, -0.7)$ with $\Gamma = ((\gamma_{jk}))_{2 \times 2} = \begin{bmatrix} 0.7 & 0.1 \\ -0.2 & 0.6 \end{bmatrix}$ and

TABLE 5 | Average estimated values and standard errors (in parenthesis) of parameters when data generated from ISOLR model for three-category case.

Parameters	$n = 500$	$n = 1000$	$n = 10,000$
$\theta_0 = -0.78$	-0.78 (0.24)	-0.77 (0.17)	-0.78 (0.05)
$\theta_1 = 0.75$	0.77 (0.23)	0.77 (0.16)	0.75 (0.05)
$\beta_1 = -0.3$	-0.31 (0.23)	-0.31 (0.16)	-0.30 (0.05)
$\beta_2 = 1.2$	1.22 (0.22)	1.21 (0.16)	1.20 (0.05)
$\beta_3 = -0.4$	-0.41 (0.18)	-0.40 (0.13)	-0.40 (0.04)
$\beta_4 = -0.7$	-0.73 (0.45)	-0.71 (0.29)	-0.70 (0.09)
$\gamma_{00} = 0.7$	0.76 (0.30)	0.74 (0.20)	0.71 (0.07)
$\gamma_{01} = 0.1$	0.12 (0.25)	0.11 (0.18)	0.10 (0.05)
$\gamma_{10} = -0.2$	-0.18 (0.27)	-0.18 (0.18)	-0.20 (0.06)
$\gamma_{11} = 0.6$	0.62 (0.23)	0.62 (0.16)	0.60 (0.05)

TABLE 6 | Average estimated values and standard errors (in parenthesis) of parameters when data generated from ISOLR model for four-category case.

Parameters	$n = 500$	$n = 1000$	$n = 10,000$
$\theta_0 = -1$	-1.01 (0.29)	-1.01 (0.20)	-1.00 (0.07)
$\theta_1 = -0.03$	-0.02 (0.25)	-0.02 (0.18)	-0.03 (0.06)
$\theta_2 = 1.5$	1.52 (0.26)	1.52 (0.19)	1.50 (0.06)
$\beta_1 = 0.3$	0.30 (0.23)	0.30 (0.16)	0.30 (0.05)
$\beta_2 = 1.4$	1.42 (0.23)	1.42 (0.16)	1.40 (0.05)
$\beta_3 = -0.3$	-0.30 (0.17)	-0.30 (0.12)	-0.30 (0.04)
$\beta_4 = -0.5$	-0.52 (0.41)	-0.52 (0.29)	-0.50 (0.09)
$\gamma_{00} = 0.3$	0.34 (0.37)	0.33 (0.26)	0.30 (0.08)
$\gamma_{01} = 0.1$	0.12 (0.37)	0.11 (0.26)	0.10 (0.08)
$\gamma_{02} = -0.3$	-0.30 (0.30)	-0.30 (0.22)	-0.30 (0.07)
$\gamma_{10} = 0.2$	0.24 (0.30)	0.23 (0.21)	0.20 (0.07)
$\gamma_{11} = 0.3$	0.32 (0.30)	0.32 (0.22)	0.30 (0.07)
$\gamma_{12} = -0.2$	-0.20 (0.25)	-0.20 (0.17)	-0.20 (0.06)
$\gamma_{20} = 0.1$	0.12 (0.31)	0.12 (0.22)	0.10 (0.07)
$\gamma_{21} = -0.2$	-0.21 (0.34)	-0.19 (0.23)	-0.20 (0.08)
$\gamma_{22} = -0.1$	-0.10 (0.26)	-0.10 (0.19)	-0.10 (0.06)

- **Four-category case:** $m = 2$ with parameters $(\theta_0, \theta_1, \theta_2, \beta_1, \beta_2, \beta_3, \beta_4) = (-1, -0.03, 1.5, 0.3, 1.4, -0.3, -0.5)$ with $\Gamma = ((\gamma_{jk}))_{3 \times 3} = \begin{bmatrix} 0.3 & 0.1 & -0.3 \\ 0.20 & 0.3 & -0.2 \\ 0.10 & -0.2 & -0.1 \end{bmatrix}$.

For each sample size, the mean and standard deviation of the MLEs obtained from these 1000 Monte Carlo replications are reported in Tables 5 and 6. Similar results are observed for the ISOLR model. As the sample size increases, the mean estimates move closer to the true parameter values, while the standard errors decrease. This further supports the empirical consistency of the proposed ISOLR model.

8.2 | Forecasting Performance

This section evaluates the forecasting performance of the proposed TSOLR and ISOLR models in comparison with the competing methods described in Section 6. Data are generated from the ISOLR model as discussed in the previous section under different parameter settings mentioned below. In the subsequent section, we further examine the robustness of the proposed models by evaluating their forecasting performance when the data are generated from two alternative models.

$$\log\left(\frac{P(Y_t \leq j)}{P(Y_t > j)}\right) = \theta_j - \left(\beta_1 S_{1t}^{(1)} + \beta_2 S_{1t}^{(2)} + \beta_3 S_{2t} + \beta_4 S_{3t}\right) - \sum_{k=0}^{m-1} \gamma_{jk} \mathbb{I}(Y_{t-1} = k). \quad (5)$$

where $j = 0, 1, \dots, m-1$, $S_{1t}^{(1)}$, $S_{1t}^{(2)}$, S_{2t} and S_{3t} are defined in Section 8.1. Here also two sets of parameter values are considered.

- **Three-category case:** $m = 2$ with parameters $(\theta_0, \theta_1, \beta_1, \beta_2, \beta_3, \beta_4) = (-0.8, 0.78, 0.2, -1.4, 0.3, 0.9)$ with $\Gamma = ((\gamma_{jk}))_{2 \times 2} = \begin{bmatrix} 0.9 & -0.2 \\ 0.1 & -0.7 \end{bmatrix}$ and
- **Four-category case:** $m = 3$ with parameters $(\theta_0, \theta_1, \theta_2, \beta_1, \beta_2, \beta_3, \beta_4) = (-0.9, 0.02, 1.2, -0.4, -1.3, 0.5, 0.7)$ with $\Gamma = ((\gamma_{jk}))_{3 \times 3} = \begin{bmatrix} -0.4 & 0.0 & 0.4 \\ -0.4 & 0.0 & 0.4 \\ -0.4 & 0.0 & 0.4 \end{bmatrix}$.

For each parameter setting, datasets of sizes $n = 500, 1000$, and $10,000$ are generated. The first 80% of each dataset is used as the training set to fit all competing models, while the remaining 20% is reserved as the test set for out-of-sample forecasting. Forecasting performance is evaluated using accuracy and weighted F1 score. The weighted F1 score is computed using the proportions of the response categories in the test set as weights. Each simulation is repeated over 1000 Monte Carlo replications and the mean and standard deviation of the performance measures are reported in Tables 7 and 8. The average computation time (in milliseconds) is also recorded for each method in the same Tables.

Results show that ISOLR achieves highest or near-highest accuracy and F1 score across all parameter sets and sample sizes. It also has low standard error of the performance measures. TSOLR and RF outperform the remaining methods in terms of both accuracy and weighted F1 score. In the four-category case, the performance differences among the models become considerably smaller. MTD and PAR have the lowest accuracy and weighted F1 score. While SVM demonstrates moderate performance in the three-category case, its performance declines in the four-category setting. LSTM and TFT require very large sample sizes to achieve competitive accuracy and F1 score. They involve very high computational time also. In both parameter settings, increasing the sample size leads to lower standard errors and marginal improvements in performance. However, the relative ranking of the methods remains largely unchanged across the three category and four category cases.

TABLE 7 | Accuracy, weighted F1 score and computation time for forecasting when data generated from ISOLR model (5) for three-category case.

Method	Accuracy (%)		F1 (%)		Time (ms)
	Mean	SD	Mean	SD	
n = 500					
TSOLR	47.00	5.18	45.70	5.28	32.31
ISOLR	48.44	4.96	46.56	5.23	28.69
MTD	41.96	5.24	35.58	7.91	4.54
PAR	39.67	5.81	24.99	5.41	15.00
RF	47.44	5.03	44.86	5.38	62.99
SVM	43.42	5.32	41.86	5.61	14.10
LSTM	41.79	5.55	29.10	8.08	424.35
TFT	43.12	6.33	36.53	8.02	4118.41
n = 1000					
TSOLR	48.11	3.54	46.72	3.75	45.56
ISOLR	49.31	3.57	47.24	3.84	38.59
MTD	42.11	3.65	36.16	6.57	8.75
PAR	40.73	3.99	24.85	3.75	29.24
RF	48.42	3.64	45.35	4.08	119.58
SVM	41.57	4.14	40.26	4.25	44.23
LSTM	44.89	4.33	35.64	7.13	808.54
TFT	45.15	4.52	39.03	6.83	7437.42
n = 10,000					
TSOLR	49.38	1.09	47.84	1.18	513.35
ISOLR	49.76	1.10	47.54	1.34	245.52
MTD	42.57	1.20	39.27	3.53	89.66
PAR	41.40	1.07	24.25	1.07	344.09
RF	49.45	1.15	45.49	1.78	1746.70
SVM	42.20	2.53	40.77	2.75	4796.19
LSTM	48.53	1.19	43.74	1.54	8062.82
TFT	49.17	1.26	46.04	2.10	66871.42

8.3 | Robustness Check

This section investigates the robustness of the proposed TSOLR and ISOLR models by generating data from alternative models. This allows us to assess whether the proposed models maintain good forecasting performance when the underlying data-generating process is misspecified. Two alternative data generating scenarios are considered: (i) data generated from a cumulative ordered probit model and (ii) data generated from a continuous latent variable model that is subsequently converted into ordinal responses.

8.3.1 | Case I: Data Generated From an Ordered Probit Model

In this scenario, ordinal categorical data are generated from a cumulative probit model (Agresti 2002). The cumulative probit model uses the inverse cumulative distribution function $\Phi^{-1}(\cdot)$ of Standard Normal distribution as its logit link function. The Ordered Probit model in this case is specified as

$$\Phi^{-1}(P(Y_t \leq j)) = \theta_j - \left(\beta_1 S_{1t}^{(1)} + \beta_2 S_{1t}^{(2)} + \beta_3 S_{2t} + \beta_4 S_{3t} \right) - \sum_{k=0}^{m-1} \gamma_{jk} \mathbb{I}(Y_{t-1} = k) \quad (6)$$

where $j = 0, 1, \dots, m-1$, $S_{1t}^{(1)}$, $S_{1t}^{(2)}$, S_{2t} and S_{3t} are defined in Section 8.1. Here also two parameter sets are considered:

- **Three-category case:** $m = 2$ with parameters $(\theta_0, \theta_1, \beta_1, \beta_2, \beta_3, \beta_4) = (-0.75, 0.78, 0.3, -1.3, 0.4, 0.7)$ with $\Gamma = ((\gamma_{jk}))_{2 \times 2} = \begin{bmatrix} 0.8 & -0.1 \\ 0.2 & -0.5 \end{bmatrix}$ and
- **Four-category case:** $m = 3$ with parameters $(\theta_0, \theta_1, \theta_2, \beta_1, \beta_2, \beta_3, \beta_4) = (-1, -0.1, 1, -0.5, -1.5, 0.7, 0.8)$ with $\Gamma = ((\gamma_{jk}))_{3 \times 3} = \begin{bmatrix} -0.3 & 0.1 & 0.2 \\ 0.3 & 0.2 & -0.4 \\ -0.1 & 0.2 & -0.1 \end{bmatrix}$.

The rest of the simulation setup is same as in Section 8.2. The resulting accuracy, weighted F1 score and computation

TABLE 8 | Accuracy, weighted F1 score and computation time for forecasting when data generated from ISOLR model (5) for four-category case.

Method	Accuracy (%)		F1 (%)		Time (ms)
	Mean	SD	Mean	SD	
n = 500					
TSOLR	42.77	5.29	35.15	5.68	37.90
ISOLR	43.61	5.16	35.88	5.67	34.08
MTD	40.48	5.34	28.38	6.03	4.84
PAR	41.01	5.17	24.22	5.16	18.41
RF	43.13	5.14	34.89	5.86	66.01
SVM	39.01	5.65	34.50	5.85	15.00
LSTM	41.07	5.00	24.28	5.07	433.63
TFT	40.17	5.87	28.19	6.26	4425.43
n = 1000					
TSOLR	43.58	3.74	35.17	4.11	50.48
ISOLR	44.26	3.80	35.73	4.25	43.32
MTD	40.85	3.86	27.89	4.60	8.41
PAR	41.09	3.69	24.04	3.68	33.17
RF	43.84	3.89	34.69	4.41	115.56
SVM	38.39	4.39	34.47	4.20	45.69
LSTM	41.11	3.61	24.40	3.78	829.84
TFT	41.64	3.80	28.67	5.23	7732.57
n = 10,000					
TSOLR	44.58	1.19	34.98	1.56	820.08
ISOLR	44.77	1.19	34.70	1.79	337.11
MTD	41.20	1.21	27.23	3.03	82.67
PAR	41.19	1.22	24.04	1.22	356.71
RF	44.61	1.21	33.46	1.79	1624.13
SVM	38.89	2.75	33.78	2.09	4494.40
LSTM	43.71	1.16	32.27	1.51	8246.82
TFT	44.24	1.23	34.26	2.08	68286.76

times are provided in Tables 9 and 10 for three-category and four-category settings respectively. The results show that ISOLR achieves highest accuracy and F1 score for both parameter setup and all sample sizes. TSOLR and RF follow closely, with performance difference becomes smaller as the sample size and number of categories increase. SVM attains moderate levels of accuracy and weighted F1 score. In contrast, MTD and PAR exhibit poor performance, particularly in terms of the weighted F1 score. For sample sizes 500 and 1000, LSTM and TFT underperform in terms of both performance measures. For $n = 10,000$, TFT attains competitive performance, but LSTM exhibits only a moderate performance. Also increasing sample size leads to lower standard deviations and marginal performance improvements without altering rankings of methods.

8.3.2 | Case II: Data Generated From a Continuous Latent Variable

In this scenario, we generate ordered categorical responses through an underlying continuous latent variable. Continuous observations Z_t are first generated from the following regression model

$$Z_t = \alpha + \beta_1 S_{1t}^{(1)} + \beta_2 S_{1t}^{(2)} + \beta_3 S_{2t} + \beta_4 S_{3t} + \gamma Z_{t-1} + \varepsilon_t \quad (7)$$

where $\varepsilon_t \sim N(0, 20^2)$ and $S_{1t}^{(1)}$, $S_{1t}^{(2)}$, S_{2t} and S_{3t} are defined in Section 8.1. As in the previous section, the following two parameter settings are considered.

- **Three-category case:** Continuous data Z_t 's are generated using the parameter set $(\alpha, \beta_1, \beta_2, \beta_3, \beta_4, \gamma) = (100, -7, 15, 20, 50, 0.5)$ in model (7). Z_t is then categorized into an ordinal response with three categories using empirical quantile-based thresholds. Let $q_{\frac{1}{3}}$ and $q_{\frac{2}{3}}$ denote the sample quantiles of Z corresponding to probabilities $\frac{1}{3}$ and $\frac{2}{3}$, respectively. The observed ordinal response Y_t is then constructed as

$$Y_t = \begin{cases} 0 & \text{for } Z_t \leq q_{\frac{1}{3}} \\ 1 & \text{for } q_{\frac{1}{3}} < Z_t \leq q_{\frac{2}{3}} \\ 2 & \text{for } Z_t > q_{\frac{2}{3}} \end{cases}$$

- **Four-category case:** Continuous data Z_t 's are generated using the parameter set $(\alpha, \beta_1, \beta_2, \beta_3, \beta_4, \gamma) = (70, 7, -15,$

TABLE 9 | Accuracy, weighted F1 score and computation time when data generated from cumulative probit model (6) for three-category case.

Method	Accuracy (%)		F1 (%)		Time (ms)
	Mean	SD	Mean	SD	
n = 500					
TSOLR	59.52	5.17	56.42	5.64	35.59
ISOLR	61.47	5.12	58.08	5.64	28.94
MTD	53.96	4.85	38.45	5.51	4.51
PAR	54.13	4.75	38.14	5.50	17.76
RF	60.83	5.29	55.65	6.01	61.88
SVM	53.72	6.08	51.30	6.16	12.04
LSTM	54.13	5.03	38.23	5.80	442.10
TFT	54.18	6.24	42.66	8.20	4116.81
n = 1000					
TSOLR	60.68	3.43	57.41	3.78	53.24
ISOLR	62.11	3.51	58.64	3.92	39.94
MTD	54.32	3.54	38.44	4.13	8.86
PAR	54.37	3.55	38.37	4.11	35.46
RF	61.58	3.73	56.17	4.31	118.21
SVM	51.20	4.83	49.53	4.58	36.60
LSTM	54.34	3.52	38.34	4.06	834.63
TFT	56.54	4.72	46.16	8.12	7442.50
n = 10,000					
TSOLR	62.02	1.16	58.60	1.30	623.86
ISOLR	62.31	1.15	59.20	1.43	262.18
MTD	54.42	1.08	38.36	1.26	88.90
PAR	54.42	1.08	38.36	1.26	403.70
RF	62.07	1.18	56.25	1.45	1702.33
SVM	51.34	3.13	49.78	2.52	3718.07
LSTM	61.07	1.14	54.77	1.46	8137.79
TFT	61.82	1.27	56.96	1.86	66904.48

16, 30, 0.4) in model (7). Z_t is then converted into an ordinal response with four categories using empirical quantile-based thresholds. Let $q_{\frac{1}{4}}$, $q_{\frac{1}{2}}$ and $q_{\frac{3}{4}}$ denote the sample quantiles of Z corresponding to probabilities $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{3}{4}$, respectively. The observed ordinal response Y_t is then constructed as

$$Y_t = \begin{cases} 0 & \text{for } Z_t \leq q_{\frac{1}{4}} \\ 1 & \text{for } q_{\frac{1}{4}} < Z_t \leq q_{\frac{1}{2}} \\ 2 & \text{for } q_{\frac{1}{2}} < Z_t \leq q_{\frac{3}{4}} \\ 3 & \text{for } Z_t > q_{\frac{3}{4}} \end{cases}.$$

The generated ordinal responses Y_t are then used to fit these models and produce forecast accuracy measures for the test set. The results are presented in Tables 11 and 12. Results show that for $n = 500$ and 1000, ISOLR achieves highest accuracy and F1 score. When $n = 10,000$, TSOLR attains highest accuracy and F1 score. In both the three-category and four-category cases, the performance measures for TSOLR and ISOLR increase as the sample size increases. Across all scenarios, TSOLR and RF shows

competitive performance, but RF shows marginally lower performance measures than ISOLR. In contrast, MTD, PAR and SVM exhibit moderate performance. Deep learning approaches (LSTM and TFT) perform poorly except for very large sample size. At $n = 10,000$, only TFT becomes competitive, but both the methods require very high computational time.

9 | Data Analysis Results

To demonstrate the practical applicability of the proposed ISOLR and TSOLR models, we first evaluate their forecasting performance using the daily AQI data for Kolkata described in Section 3. To further examine the generalizability of the proposed models, we also apply them to the daily AQI data for Mumbai.

9.1 | Daily AQI of Kolkata

To examine the temporal dependence in the AQI series and identify an appropriate model order, we compute several lag-wise

TABLE 10 | Accuracy, weighted F1 score and computation time when data generated from cumulative probit model (6) for four-category case.

Method	Accuracy (%)		F1 (%)		Time (ms)
	Mean	SD	Mean	SD	
n = 500					
TSOLR	49.06	5.03	48.19	5.23	50.88
ISOLR	50.93	4.90	49.84	5.14	40.23
MTD	43.48	5.26	39.12	5.87	5.47
PAR	34.17	4.86	24.43	6.81	20.44
RF	49.33	5.16	47.47	5.67	79.56
SVM	43.84	5.46	42.29	5.71	15.09
LSTM	39.65	5.81	29.10	7.81	452.18
TFT	39.97	7.22	32.81	8.67	4417.97
n = 1000					
TSOLR	49.99	3.64	49.21	3.75	97.48
ISOLR	51.16	3.53	50.14	3.66	60.47
MTD	43.48	3.55	39.60	4.19	10.55
PAR	34.09	3.46	24.55	5.94	40.05
RF	50.03	3.55	48.49	3.89	153.62
SVM	41.76	4.12	40.49	4.29	47.80
LSTM	41.23	3.71	31.61	4.37	861.29
TFT	43.55	4.84	37.59	6.76	7766.95
n = 10,000					
TSOLR	51.29	1.16	50.63	1.20	1281.50
ISOLR	51.66	1.16	50.75	1.21	419.24
MTD	44.06	1.18	41.18	1.38	101.75
PAR	34.06	1.18	25.68	3.34	422.17
RF	51.03	1.20	50.02	1.39	1919.92
SVM	42.87	2.50	40.95	2.85	4471.18
LSTM	47.97	1.22	46.70	1.32	7982.28
TFT	50.58	1.40	49.35	1.69	67457.69

dependency measures described in Table 2. Figure 10 shows that the strength of dependence decreases steadily as the lag increases. This pattern suggests that the dependence is mainly captured by the first lag and hence a lag value Y_{t-1} is included in the subsequent analysis.

To evaluate the forecasting performance of the models, the data from 2019 to 2023 are used as the training set, while the data from 2024 are used as the test set for out-of-sample forecasting. We compare the models described in Sections 4 and 6 using the accuracy and weighted F1 score. Each model is trained to predict the AQI category of the next day using the current AQI state and seasonal information. The ISOLR model incorporates discrete seasonal indicators in the linear predictor, as specified in model (3). This makes the model easy to interpret while capturing the effects of major seasonal patterns. In contrast, the TSOLR model uses sinusoidal terms to represent seasonality, as described in model (2). This approach provides a smooth representation of seasonal variation and captures both weekly and annual cycles, including higher-frequency seasonal events such as Diwali.

To complement the proposed models, we also consider the classical time series models MTD and PAR. These models do not incorporate external covariates such as seasonal effects and their forecasts are based only on the lag-1 AQI category. In addition, we implement several machine learning and deep learning models, including Random Forest (RF), Support Vector Machine (SVM), Long Short-Term Memory (LSTM) and the Temporal Fusion Transformer (TFT). These models are trained using the seasonal indicator variables together with the lag-1 AQI category. Although they are not specifically designed for ordinal responses, they can capture complex non-linear relationships and therefore serve as useful benchmark methods. Finally, to examine whether forecasting the original continuous AQI values can improve prediction by making use of the more informative raw measurements, we also fitted the TBATS and SARIMA models to the continuous AQI series.

The forecasting results for the Kolkata AQI data are presented in Table 13. Among all the methods, the TSOLR model achieves the best overall performance, with an accuracy of 73.50% and

TABLE 11 | Accuracy, weighted F1 score and computation time when data generated continuously from model (7) for three-category case.

Method	Accuracy (%)		F1 (%)		Time (ms)
	Mean	SD	Mean	SD	
n = 500					
TSOLR	60.83	5.24	60.68	5.34	39.83
ISOLR	63.29	5.17	63.07	5.24	40.82
MTD	57.80	5.30	57.74	5.48	4.58
PAR	57.84	5.24	57.85	5.24	17.73
RF	61.53	5.49	59.88	6.16	63.76
SVM	57.25	5.73	55.87	6.16	11.56
LSTM	52.22	7.82	44.90	10.13	458.40
TFT	51.46	8.78	47.15	10.41	4104.39
n = 1000					
TSOLR	62.63	4.02	62.36	4.11	66.13
ISOLR	63.67	3.95	63.48	4.02	55.38
MTD	57.99	3.88	57.99	3.88	9.28
PAR	57.99	3.88	57.99	3.88	35.17
RF	62.63	4.08	61.10	4.60	126.06
SVM	55.54	4.27	54.57	4.35	35.56
LSTM	57.07	4.55	49.16	6.68	843.74
TFT	57.35	5.65	53.32	7.29	7431.45
n = 10,000					
TSOLR	64.33	1.13	63.93	1.15	841.94
ISOLR	63.79	1.12	63.54	1.17	292.06
MTD	57.84	1.18	57.84	1.18	88.03
PAR	57.84	1.18	57.84	1.18	378.45
RF	63.39	1.20	62.26	1.49	1756.20
SVM	55.88	2.32	54.70	1.86	3312.45
LSTM	60.59	1.18	60.22	1.43	8080.42
TFT	63.66	1.22	62.95	1.47	66894.18

a weighted F1 score of 73.54%. The ISOLR model ranks second, with an accuracy of 72.95% and a weighted F1 score of 72.96%. Among the remaining methods, MTD(1), PAR(1) and SVM achieve identical performance, each with an accuracy and weighted F1 score of 72.68%. LSTM performs similarly, with only a slight reduction in both measures. RF achieves a slightly lower accuracy of 71.31% and a weighted F1 score of 71.38%. In contrast, TFT shows noticeably poorer performance, with an accuracy of 68.22% and a weighted F1 score of 67.07%. The continuous AQI forecasting methods, SARIMA and TBATS, perform considerably worse because they first predict continuous AQI values and then convert them into categories. As a result, even small forecasting errors near category boundaries can lead to incorrect category predictions. Figure 11 compares the observed AQI categories in the test set with the forecasts obtained from the different models.

9.2 | Daily AQI of Mumbai

In this section, we consider the categorical AQI data for Mumbai and analyze it with the above models. Figure 12a,b shows the

original continuous AQI series and the corresponding categorical AQI series, respectively. Both figures reveal clear seasonal patterns throughout the study period. The “Very Poor” AQI category was observed only once, in 2022. In addition, only a few “Poor” AQI days were recorded during the last two winter seasons, suggesting an improvement in air quality in recent years. Exploratory data analysis results are presented in the appendix for brevity.

To examine the serial dependence and identify an appropriate model order, we compute several lag-wise dependence measures described in Table 2. Figure 13 shows that the strength of dependence decreases steadily as the lag increases. This suggests that most of the temporal dependence is captured by the first lag and supports the use of a first-order type Markov model.

The Mumbai AQI data are divided into a training set and a test set. The observations from 2019 to 2023 are used as the training set, while the observations from 2024 are used as the test set for out-of-sample forecasting. All the models are fitted using the

TABLE 12 | Accuracy, weighted F1 score and computation time when data generated continuously from model (7) for four-category case.

Method	Accuracy (%)		F1 (%)		Time (ms)
	Mean	SD	Mean	SD	
n = 500					
TSOLR	46.73	5.26	45.94	5.46	44.76
ISOLR	48.33	5.09	47.25	5.30	37.87
MTD	42.18	5.25	38.14	7.18	4.42
PAR	43.57	5.24	43.57	5.23	17.58
RF	46.98	5.32	43.45	6.12	64.72
SVM	44.13	5.60	42.67	5.85	13.05
LSTM	38.28	6.72	28.88	7.94	472.04
TFT	37.85	7.86	31.69	8.40	4409.81
n = 1000					
TSOLR	48.02	3.71	47.18	3.87	71.88
ISOLR	49.03	3.74	47.85	3.89	51.87
MTD	42.52	3.86	39.51	6.13	8.67
PAR	43.47	3.63	43.47	3.63	34.37
RF	47.66	3.95	44.03	4.60	124.20
SVM	41.60	4.81	40.73	4.93	42.28
LSTM	44.01	4.24	34.31	5.92	861.88
TFT	42.80	5.34	36.23	6.55	7727.46
n = 10,000					
TSOLR	49.73	1.19	48.59	1.23	1259.63
ISOLR	49.71	1.18	48.23	1.27	404.78
MTD	43.59	1.22	43.53	1.46	82.97
PAR	43.60	1.20	43.60	1.21	353.88
RF	48.96	1.29	45.71	1.86	1686.46
SVM	41.66	3.50	40.67	3.40	3865.09
LSTM	47.01	1.22	44.07	1.63	8290.82
TFT	49.42	1.22	47.52	1.53	68343.26

training data and their forecasting performance on the test set is summarized in Table 14.

Among all the methods, the TSOLR model achieves the best overall performance, with the highest accuracy of 82.79% and a weighted F1 score of 82.46%. The RF model performs nearly as well, achieving an accuracy of 82.24% and a weighted F1 score of 82.00%. The ISOLR model also performs well, with an accuracy of 81.97% and a weighted F1 score of 81.76%. Among the remaining machine learning and deep learning methods, TFT, LSTM and SVM show comparable performance, with accuracies ranging from 81.15% to 81.64% and weighted F1 scores above 80.9%. The classical models MTD(1) and PAR(1) perform slightly worse, each achieving an accuracy and weighted F1 score of 80.87%. The continuous AQI forecasting methods, SARIMA and TBATS, perform considerably worse because they first forecast continuous AQI values and then convert them into categories. As a result, even small forecasting errors near category boundaries can lead to incorrect category predictions. Figure 14 compares the observed AQI categories in

the 2024 test set with the forecasts obtained from the different models.

10 | Discussion

This study proposes two ordinal logistic regression models, TSOLR and ISOLR, for modelling multiple seasonality in categorical Air Quality Index (AQI) data. Motivated by the TBATS model (Livera et al. 2011), which uses trigonometric Fourier series to represent complex seasonal patterns in continuous time series, the proposed TSOLR model incorporates Fourier seasonal terms into an ordinal logistic regression framework to capture smooth seasonal variation. The ISOLR model provides a complementary approach by replacing the Fourier terms with seasonal indicator variables, allowing it to capture discrete seasonal changes. Together, these two models offer a flexible framework for modeling both smooth and discrete seasonal patterns in categorical AQI data. Simulation studies and real data analyses demonstrate that the proposed models provide accurate, robust and

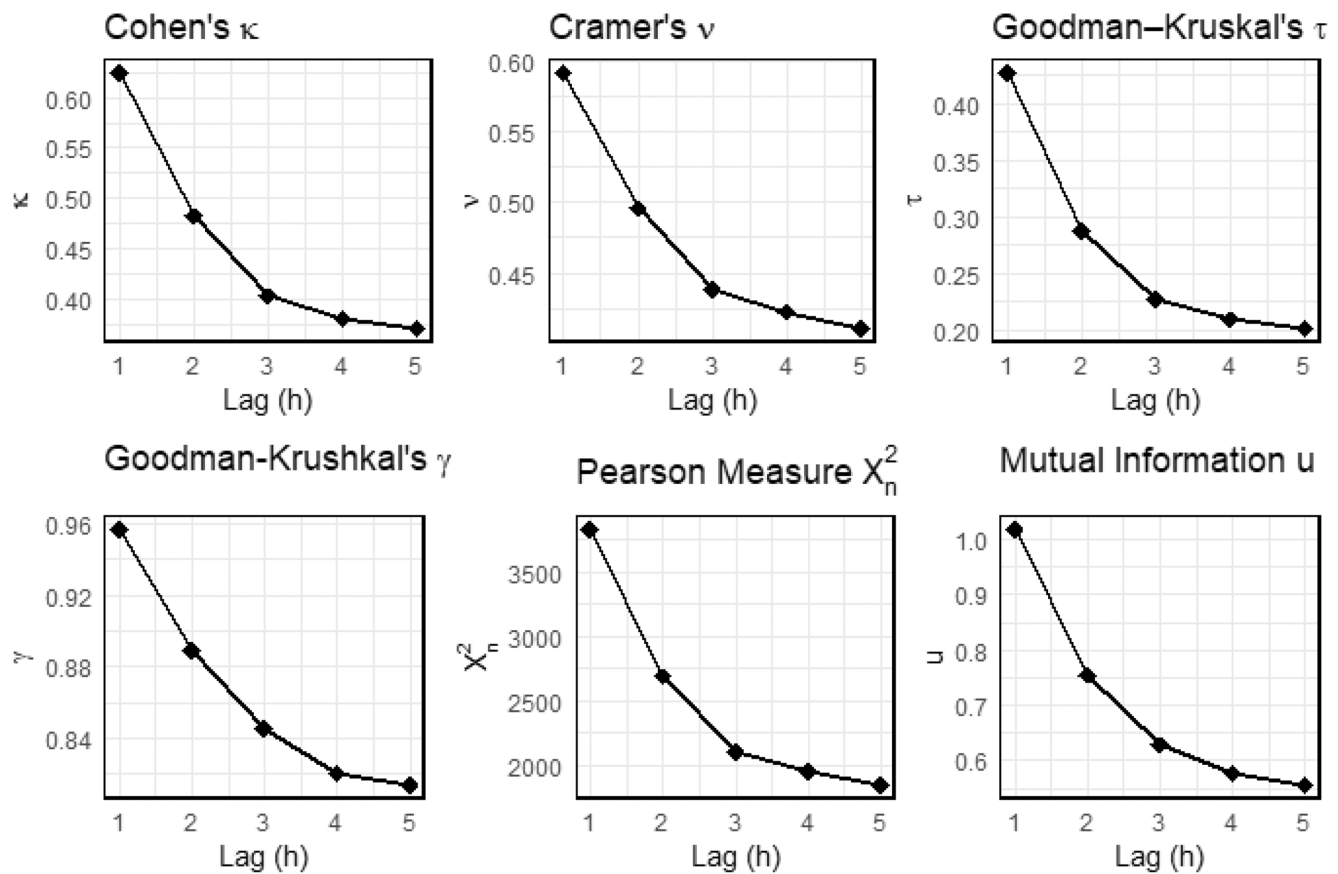


FIGURE 10 | Auto-association measures of the categorical AQI time series computed over varying lag values for Kolkata.

TABLE 13 | Forecasting results for Kolkata data analysis.

Models	Accuracy	Weighted F1
TSOLR	73.50%	73.54%
ISOLR	72.95%	72.96%
MTD(1)	72.68%	72.68%
PAR(1)	72.68%	72.68%
RF	71.31%	71.38%
SVM	72.68%	72.68%
LSTM	72.40%	72.43%
TFT	68.22%	67.07%
SARIMA	25.14%	10.10%
TBATS	51.91%	47.96%

computationally efficient forecasts, making them effective tools for forecasting ordinal AQI categories.

The two models have complementary strengths. The ISOLR model is easy to interpret because it directly estimates the effects of seasonal factors such as different seasons, weekends and festival periods like Diwali. In contrast, the TSOLR model provides a smooth representation of seasonal variation and can effectively capture annual and weekly cycles, as well as short-term fluctuations associated with special events.

Extensive simulation studies demonstrate that both proposed models produce consistent parameter estimates and accurate

forecasts. They perform particularly well when the underlying data follow an ordinal or latent variable structure and their forecasting performance remains stable under alternative data-generating mechanisms. Compared with the classical categorical time series models MTD and PAR, the proposed models achieve substantially better forecasting accuracy because they can incorporate seasonal information. Machine learning methods, such as Random Forest and SVM, provide competitive results in some settings but generally do not outperform the proposed models. Although deep learning methods, especially TFT, become competitive for very large sample sizes, they require considerably more computational time. It should also be noted that the smaller sample sizes, while adequate for statistical models, may not be sufficient for training deep learning models effectively.

The real data analyses using the Kolkata and Mumbai AQI datasets further support the usefulness of the proposed methods. In both applications, TSOLR and ISOLR are among the best-performing models in terms of forecasting accuracy and weighted F1 score. The classical Markov chain models MTD and PAR consistently perform worse because they rely only on past AQI categories and cannot incorporate seasonal covariates. Similarly, continuous time series models such as TBATS and SARIMA do not perform well after converting continuous forecasts into AQI categories, since even small numerical prediction errors near category boundaries can lead to incorrect classifications. Although the improvements in forecasting accuracy over some competing methods are modest, the proposed models achieve

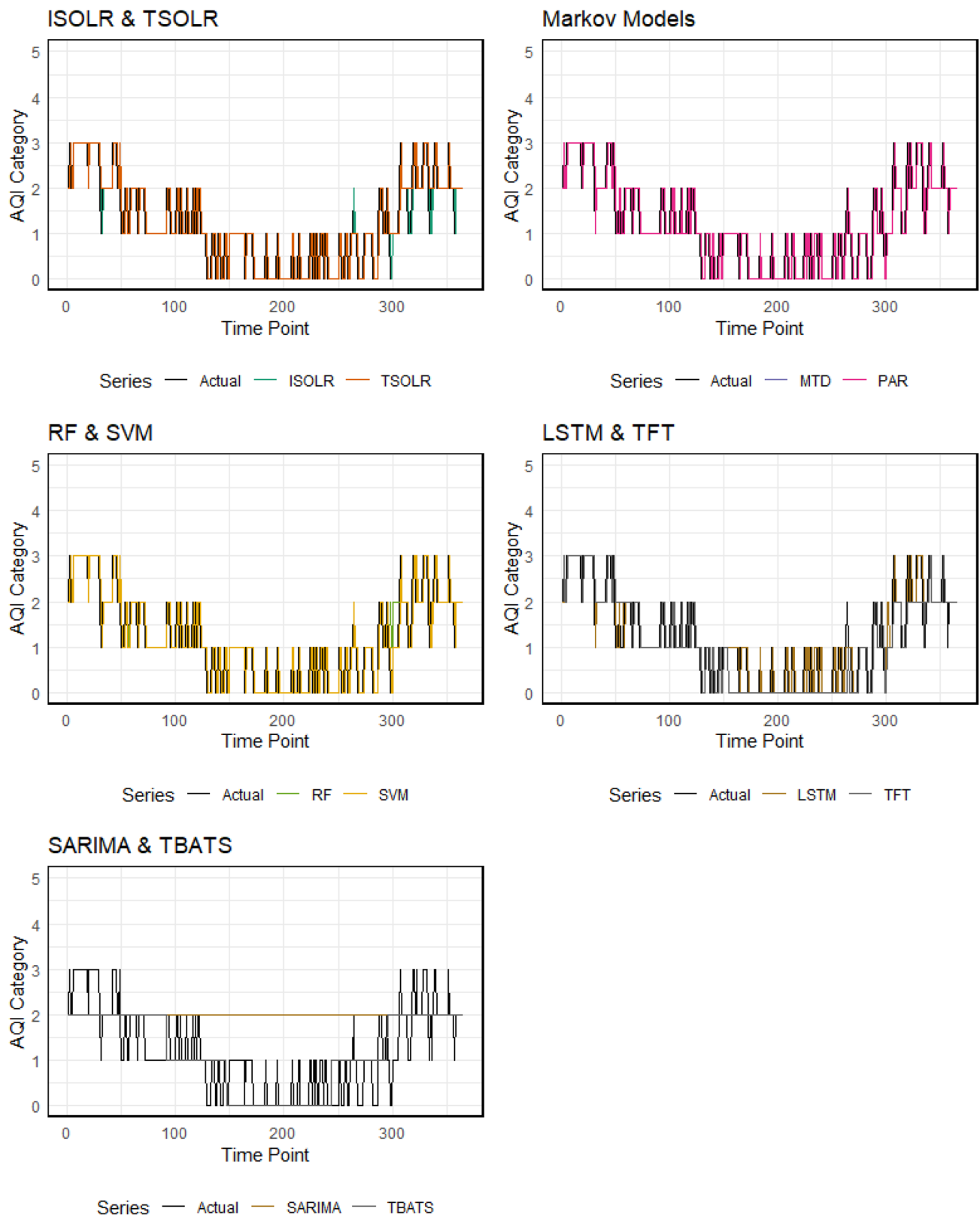


FIGURE 11 | Predicted AQI categories for various models along with the true categories for Kolkata dataset.

these results with substantially lower computational cost, making them attractive for practical applications.

There are several directions for future research. The proposed models can be extended to account for spatial dependence

by jointly modeling AQI data from multiple monitoring stations. Their predictive performance may also be improved by including additional meteorological and pollution-related variables, such as wind speed, humidity and particulate matter concentrations ($PM_{2.5}$ and PM_{10}). Relaxing the proportional

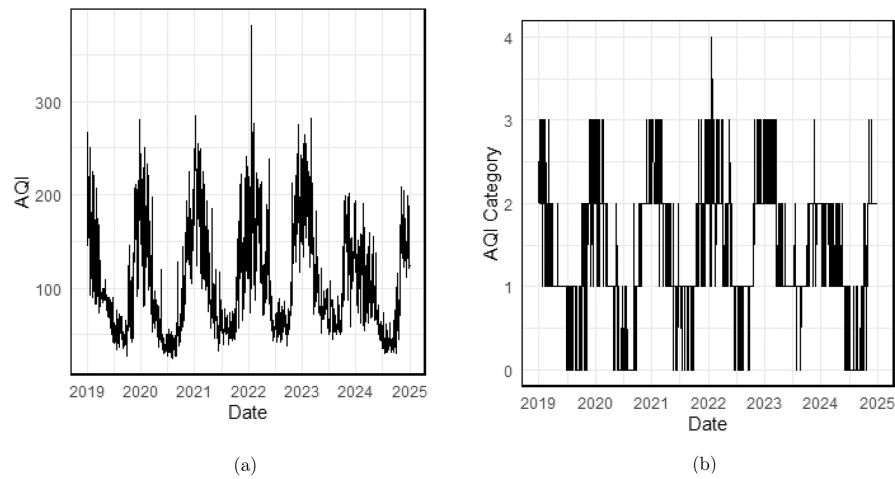


FIGURE 12 | Daily Air Quality Index (AQI) time series (original and categorical) of Mumbai during 2019–2024 showing the variation in air pollution levels over time with noticeable seasonal and episodic fluctuations. (a) Daily Air Quality Index (AQI) time series of Mumbai during 2019–2024 (higher values indicate poorer air quality), (b) Daily categorical AQI time series for Mumbai from 2019 to 2024.

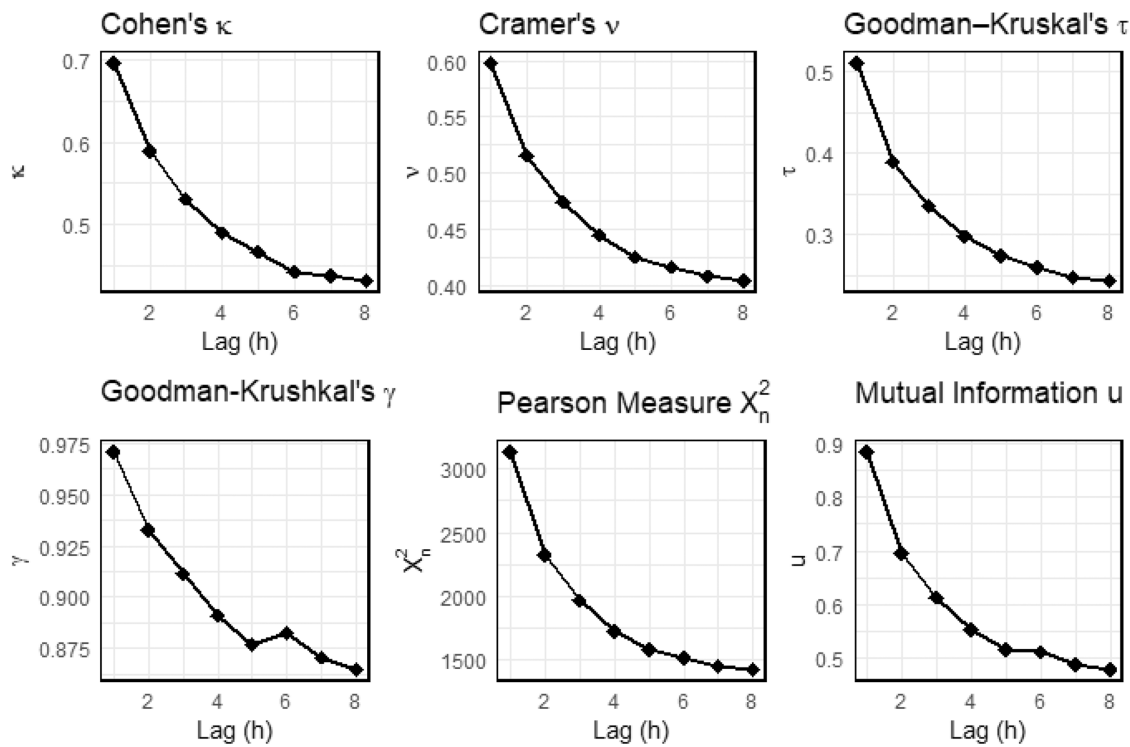


FIGURE 13 | Auto-association measures of the categorical AQI time series computed over varying lag values for Mumbai.

TABLE 14 | Forecasting results for Mumbai data analysis.

Models	Accuracy	Weighted F1
TSOLR	82.79%	82.46%
ISOLR	81.97%	81.76%
MTD(1)	80.87%	80.87%
PAR(1)	80.87%	80.87%
RF	82.24%	82.00%
SVM	81.15%	80.97%
LSTM	81.42%	81.19%
TFT	81.64%	81.22%
SARIMA	39.62%	22.48%
TBATS	39.62%	22.48%

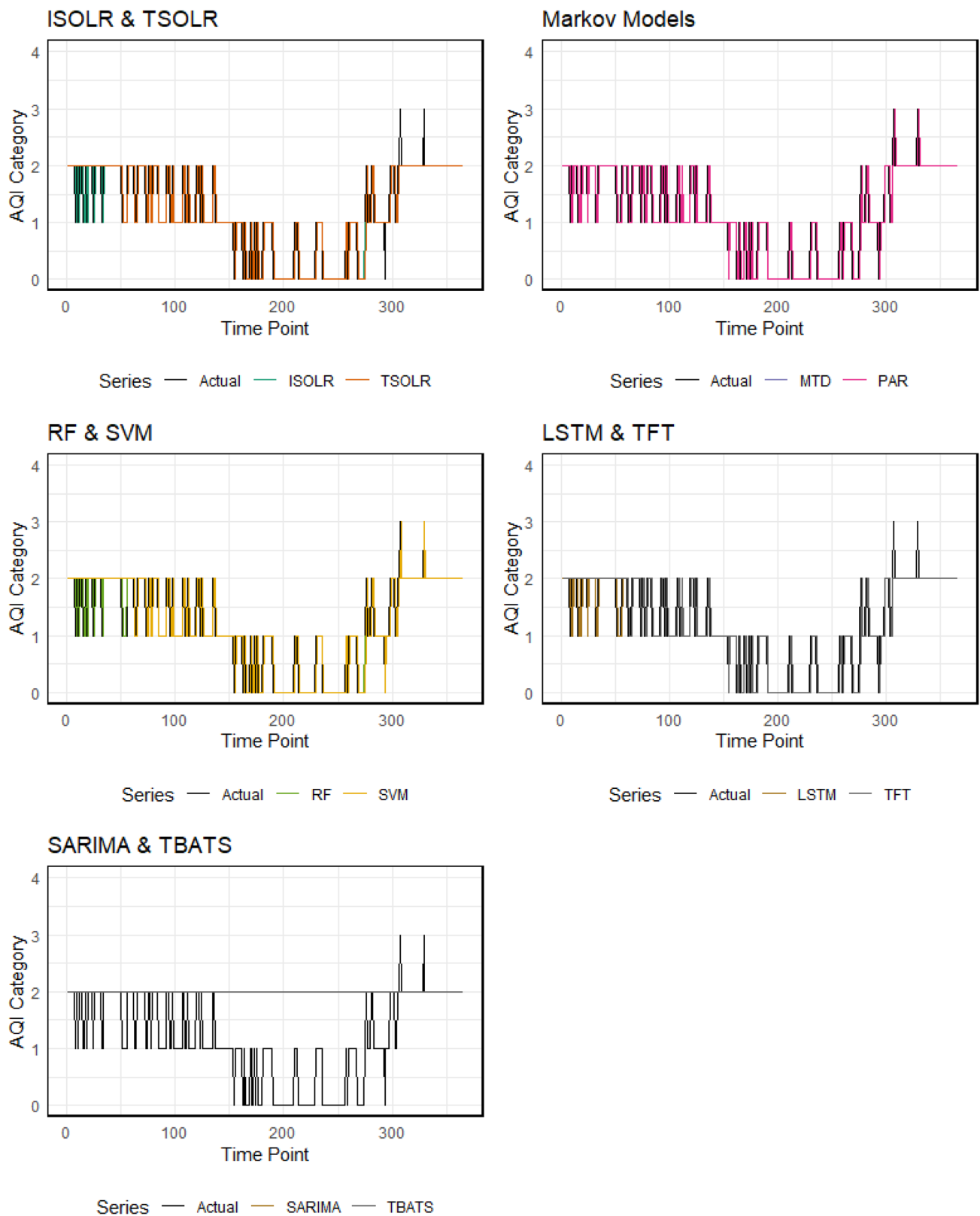


FIGURE 14 | Predicted AQI categories for various models along with the true categories for Mumbai dataset.

odds assumption through partial or non-proportional odds models for seasonal coefficients would provide greater flexibility when seasonal effects differ across AQI categories. Future work may also investigate uncertainty in forecasting performance using resampling methods designed for dependent time

series, such as block bootstrap or rolling-window evaluation. Finally, Bayesian versions of the proposed models and their integration into real-time air quality monitoring and decision support systems offer promising directions for further development.

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The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available in GitHub at <https://github.com/ganirban004/aqi-logistic-paper.git>. These data were derived from the website of Central Pollution Control Board, India (<https://airquality.cpcb.gov.in/ccr/#/caaqm-dashboard-all/caaqm-landing/aqi-repository>), which is publicly available.

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Appendix A

Exploratory Analysis of Mumbai Data

Here the results of exploratory analysis of categorical AQI data for Mumbai is shown. Figure A1a shows the trends of various categories across various years. It shows a clear improvement of air quality across years.

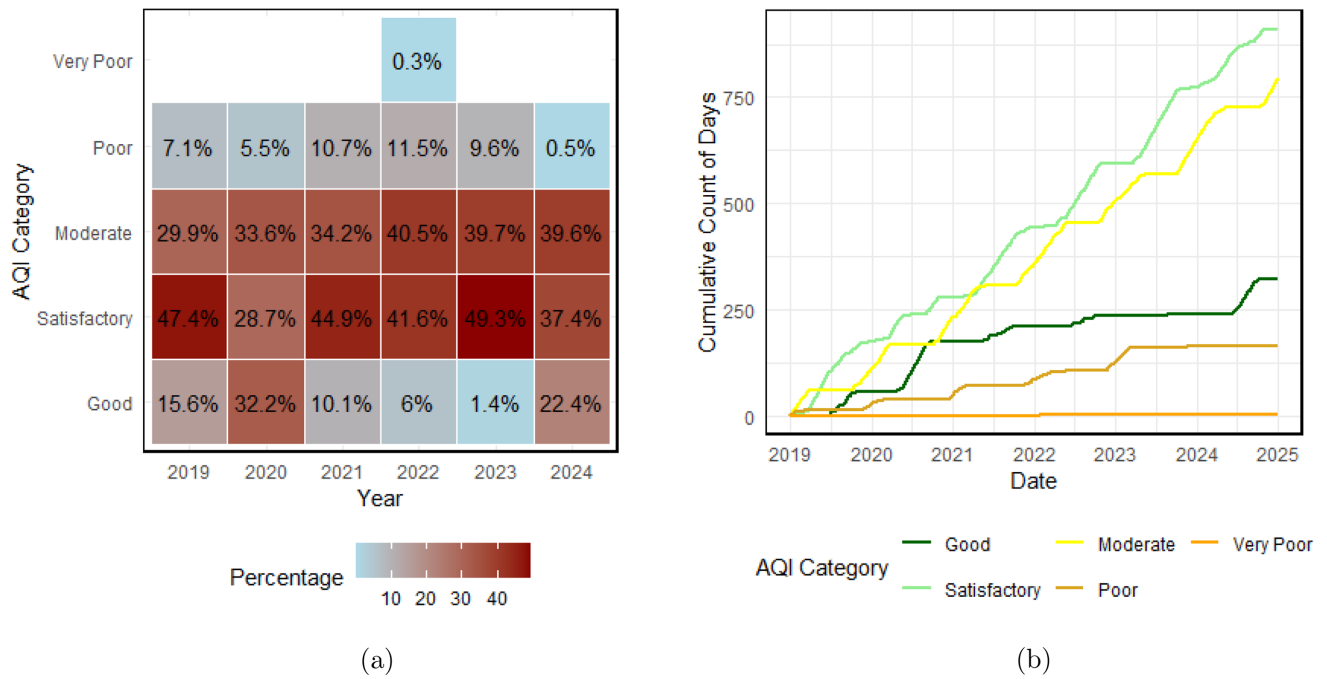


FIGURE A1 | Yearly trend and rate evolution graph of categorical AQI levels in Mumbai from 2019-2024. (a) Yearly Trend of categorical AQI levels in Mumbai from 2019 to 2024, (b) Daily rate evolution graph of categorical AQI levels in Mumbai from 2019-2024.

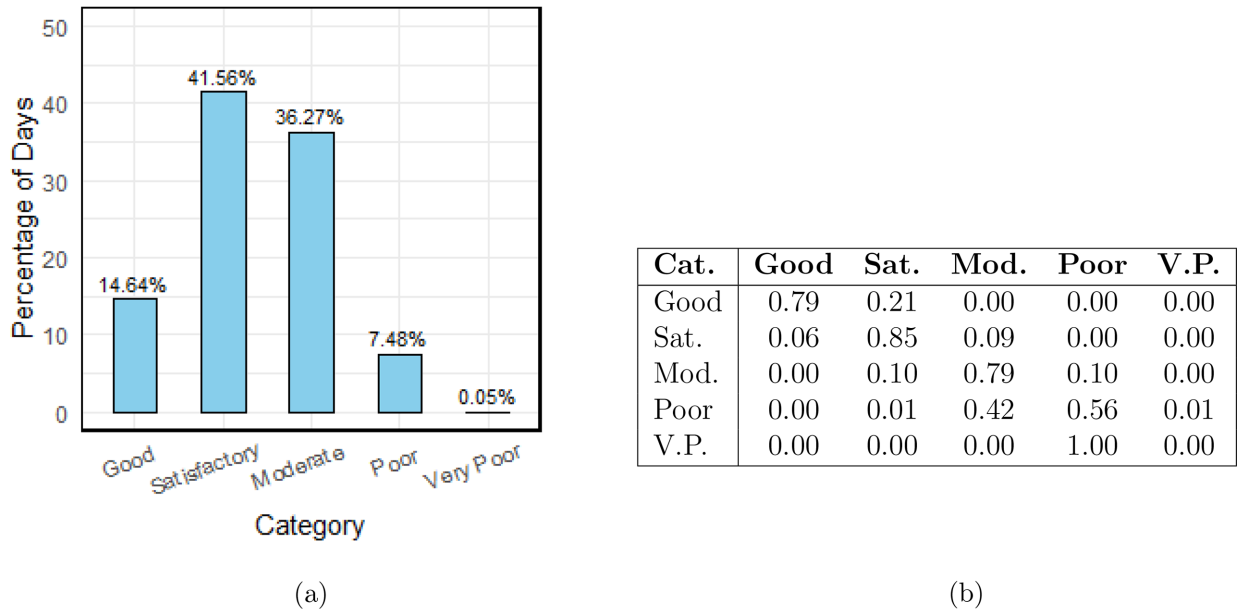


FIGURE A2 | Frequency distribution of AQI categories and Transition probability matrix for the categorical AQI time series. (a) Frequency distribution of AQI categories in Mumbai over the entire study period (2019–2024), (b) Transition Probability Matrix of Air Quality Categories over the entire study period (2019-2024).

A sudden increase for “Good” category in 2020 is due to the COVID-19 lockdown. Figure A1b shows the rate evolution graph for Mumbai data. Figure A2a indicates that the “Satisfactory” category is the most prevalent, followed by the “Moderate” category. The transition probability matrix in Figure A2b indicates strong temporal persistence within the same category. The month-category heatmap in Figure A3 reveals a clear seasonal pattern in air quality, with comparatively poorer air quality observed during the winter months and substantial improve-

ment during the Monsoon season. Season-wise, weekly and festival-wise variations are illustrated in Figures A4, A5 and A6 respectively. Figure A7 shows the yearly trend in AQI categories during Diwali period. It suggest that the adverse impact of Diwali on air quality has been increasing over the years, which is a dangerous sign from both environmental and public health perspectives. Overall, the categorical AQI data for Mumbai exhibit multiple seasonal patterns, similar to those observed for Kolkata.

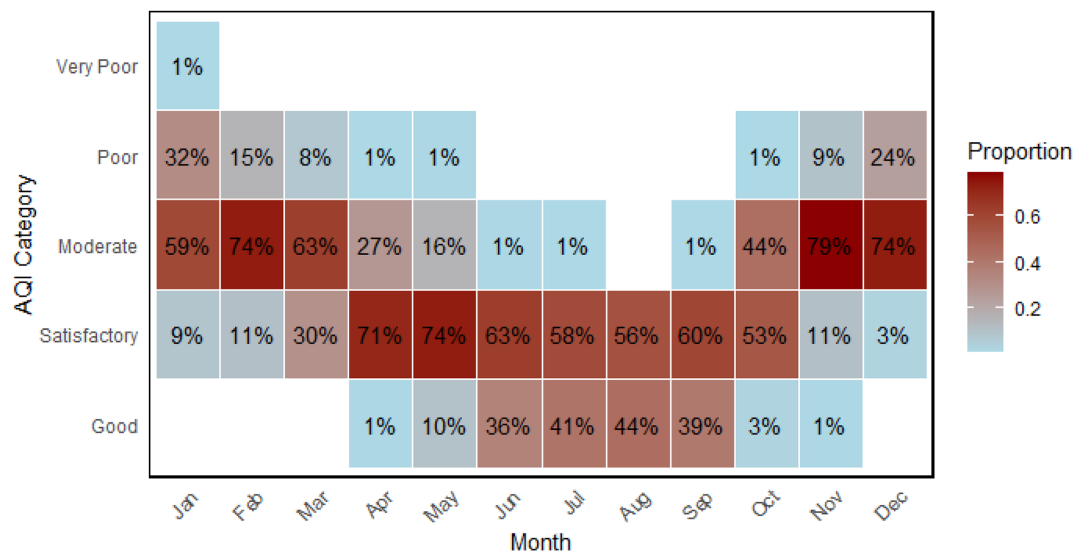


FIGURE A3 | Month-category heatmap of AQI categories using intensity measure.

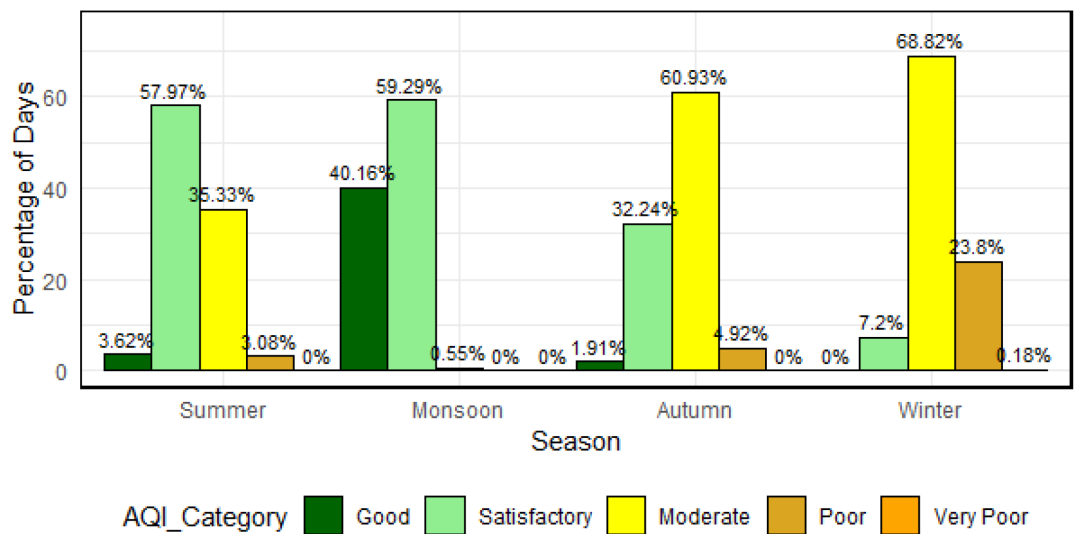


FIGURE A4 | Seasonal distribution of categorical AQI levels in Mumbai from 2019 to 2024.

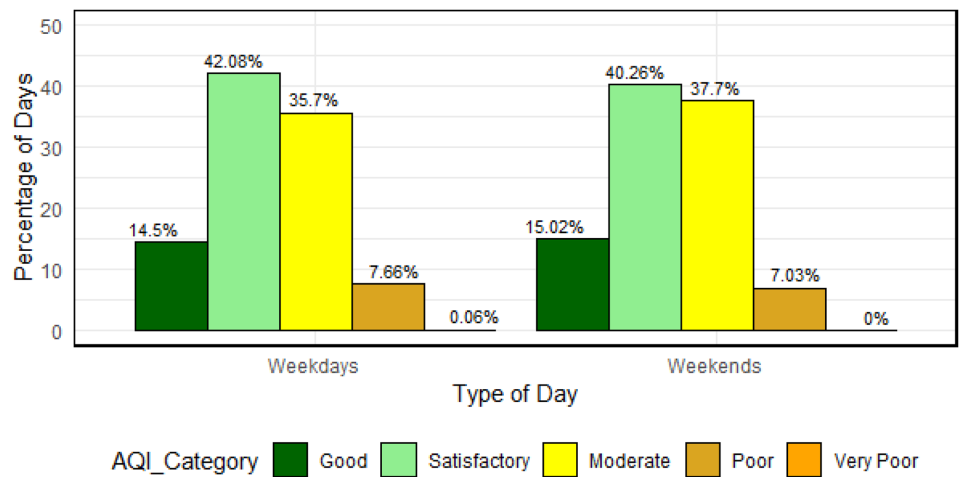


FIGURE A5 | Frequency distribution of AQI categories during weekdays and weekends in Mumbai (2019-2024).

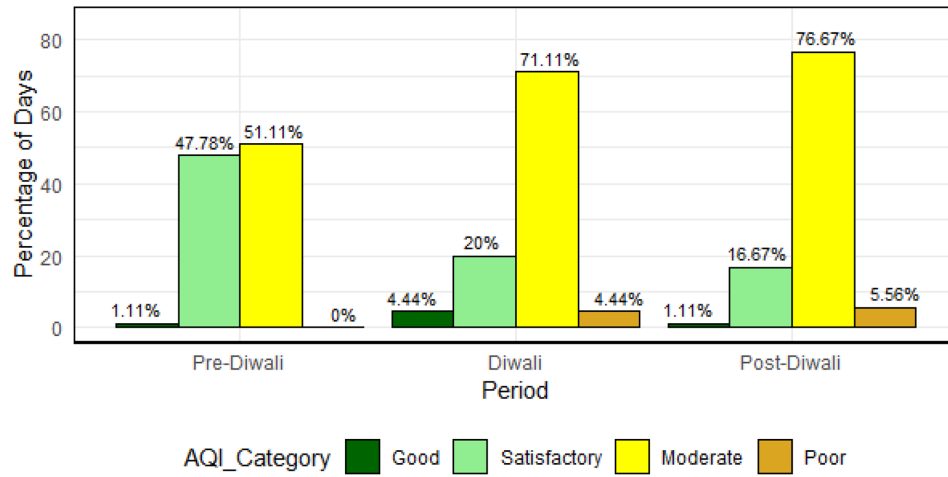


FIGURE A6 | Frequency distribution of AQI categories during the Pre-Diwali, Diwali and Post-Diwali periods in Mumbai (2019-2024).

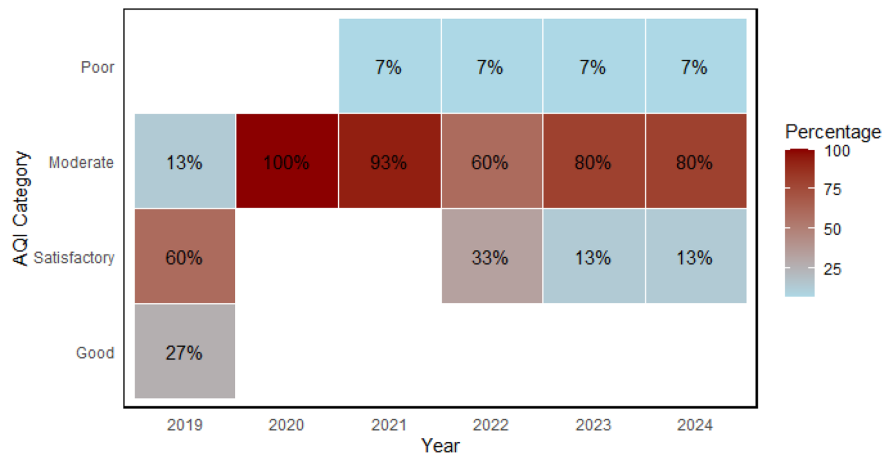


FIGURE A7 | Yearly trend of AQI categories during the Diwali period in Mumbai (2019-2024).

TABLE A1 | AIC values for different combinations of K_1 and K_2 for AQI data of Kolkata.

K_1	K_2		
	1	2	3
1	2618.44	2620.33	2623.51
2	2590.92	2592.90	2596.00
3	2593.24	2595.21	2598.30
4	2591.05	2592.98	2596.06
5	2584.40	2586.34	2589.43
6	2586.69	2588.65	2591.76
7	2590.67	2592.63	2595.74
8	2594.20	2596.14	2599.25
9	2598.12	2600.07	2603.18
10	2602.00	2603.93	2607.05

Note: The value in bold font indicates the minimum AIC.

Choice of K_i in TSOLR

Here the optimal value of K_i is chosen by minimizing the AIC over a set of candidate values. The maximum possible value of K_i is $\left\lfloor \frac{P_i}{2} \right\rfloor$, where P_i is the seasonal frequency of i^{th} seasonality. If $K_i > \left\lfloor \frac{P_i}{2} \right\rfloor$, then the design matrix will be singular. Livera et al. (2011) adopted the approach of setting $K_i = \left\lfloor \frac{P_i}{2} \right\rfloor$. However they also noted that most of the seasonal

TABLE A2 | AIC values for different combinations of K_1 and K_2 for AQI data of Mumbai.

K_1	K_2		
	1	2	3
1	2012.52	2015.76	2015.96
2	2007.18	2010.42	2010.75
3	2009.36	2012.58	2012.88
4	2011.22	2014.45	2014.73
5	2008.70	2011.91	2012.28
6	2009.81	2013.00	2013.38
7	2011.53	2014.73	2015.15
8	2013.76	2016.96	2017.37
9	2017.58	2020.77	2021.17
10	2016.45	2019.64	2020.03

Note: The value in bold font indicates the minimum AIC.

components require fewer harmonics. Our study also shows similar pattern. Tables A1 and A2 report the AIC values for different combinations of K_1 and K_2 for AQI data of Kolkata and Mumbai, respectively. Here $P_1 = 365$ and $P_2 = 7$ and so $K_1 \in 1, 2, \dots, 10$ and $K_2 \in \{1, 2, 3\}$. The results provide sufficient evidence for restricting the search space over the set $\{1, 2, \dots, \min(10, \left\lfloor \frac{P_1}{2} \right\rfloor)\}$.